

PHARMACEUTICS I LAB MANUAL

TABLE OF CONTENTS

1. Pharmaceutical Measurements.....	2
2. Interpretation of Prescriptions	8
3. Formulation of liquid Dosage Forms, Aromatic Water.....	10
4. Spirits, Tinctures and Mixtures.....	17
5. Syrups, Elixirs and Linctuses.....	23
6. Solutions Used in the Mouth: Gargles, Mouthwashes, Throat Paints...	31
7. Solutions Instilled into Body Cavities: Douches, Enemas, EarDrops, Nasal Drops.....	35
8. Solutions for External Use: Collodions, Liniments, Lotions, Paints.....	44
9. Emulsions.....	51
10. Suspension.....	65

EXPERIMENT I

Pharmaceutical Measurements

Objective: To introduce students to pharmaceutical measurements.

Pharmaceutical preparations, as the final forms used for medication, must be effective and safe. These requirements demand:

- Quantitative accuracy and
- Qualitative accuracy

1) Qualitative Accuracy

- Involves assuring the identity of ingredients by constant vigilance.

It requires:

- (a) Reading labels carefully and doing this three times: when the container is taken from the shelf, when it is used, and after it has been used.
- (b) Observing the appearance such as the color, shape and texture as well as the odor of every ingredient.

2) Quantitative Accuracy

- Involves confirming the amount of the ingredients to be exact.
- Quantitative accuracy may be achieved by carefully
 - (a) Choosing the instrument (balance or measure) to be used in relation to: the amount of the ingredient, the nature of the ingredient, and the degree of accuracy desired, etc.
 - (b) Reading the markings corresponding to the volume or weight measured.

- Accuracy of measurement, although important in either case, is more important especially in hospital and community pharmacies than in pharmaceutical factories. This is because the latter involve frequent supervision and therefore there is more chance for detecting failure of accuracy in measurement.
- Knowledge and application of accurate pharmaceutical measurements are essential for pharmacists to practice their profession.

Pharmaceutical measurements mostly involve: Weighing and Measuring Volumes.

Weight

- Equally as important as the balance in the weighing process are the mass standards, the weights.
- Good weights are commonly made of polished brass and to increase resistance to corrosion they may be plated with Nickel, Chromium, Gold or Platinum.
- Weights of high accuracy are also made of stainless alloys.
- Small weights of less than 1g (500 mg to 50 mg) are made of Tantalum and Aluminum for less than 50mg.

Cares that should be taken in weighing

- Good technique in the weighing process is important to both accuracy and prolonged usefulness of the equipment. Some of the rules that can serve as guides are as follows:
 1. Keep the balance in a well-lighted place, as free as possible from vibration, dust, moisture, corrosive vapors, wind, etc.
 2. Keep the balance cover down except when it is in use.

3. Keep the balance clean at all times (remove by a soft brush or a dry clean towel).
4. Use powder papers, and watch glasses or stoppered bottles (for corrosive substances) to prevent direct contact of the pan(s) and the material to be weighted.
5. Support the pan(s) at all times except when equilibrium is tested. Weights and materials should never be added to or removed from the pans unless the pans are supported.
6. Handle weights only with forceps because fingerprints increase their weights or accelerate corrosion.
7. Adjust the scale to perfect balance (Zero position) before each weighing. Failure to adjust the scale is the most common source of error.
8. Clean weights with a soft clean towel (when required) and if this is not sufficient, use a paste of Precipitate Calcium Carbonate and Glycerine followed by cleaning with a moistened towel and drying.

Weighing Techniques for Torsion Balances

1. Adjust the balance by means of the leveling screws so that the index pointer is at zero.
 2. With the balance arrested, open the balance lid (cover) and place the desired weights on the right pan and the material to be weighed on the left pan.
 3. Unlock the balance to observe if too little or too much material was deposited.
 4. Using a spatula, remove or add material, arresting the balance each time a transfer is made.
 5. When equilibrium is established, the balance should be arrested, the lid closed, and the arrest released to check the equilibrium.
- In dispensing (Hospital and Community Pharmacies), the type of balance used is called prescription Balance.
 - There are two main types of Prescription Balances:

1) Class A Prescription Balance e.g. Torsion Prescription balance

2) Class B Prescription Balance

The major purpose being the same (i.e. weighing), balances are differentiated from each other by the following parameters:

- 1) **Sensitivity:** Is the smallest weight that makes a perceptible change in the pointer of a balance, which indicates the equilibrium position.
- 2) **Sensitivity Requirement (SR):** Is the weight necessary to move the pointer of the balance one division on the index plate against which equilibrium is observed.
 - SR for class A Prescription balance is 6 mg while that of class B prescription balance is 30 mg.
- 3) **Capacity:** Is the maximum weight which a balance can weigh.

The capacity of most Class A and Class B Prescription Balance is 120g.

In order to avoid errors of 5% or more, the minimum amount that can be weighted on Class A Prescription Balance is 120mg and on Class B Prescription Balance is 600 mg.

$$\text{Smallest Weight that can be weighted} = \frac{\text{SR} \times 100}{\text{Permissible Percent of Error}}$$

Major parts of a Balance:

Pan(s),	Spirit Level,
Balance Pointer,	Screw Feet,
Index Plate,	Balance Cover,
Beam(s),	Pan Support.

Determination of Sensitivity Requirement for Class A Prescription Balance

1. Level the balance and determine the rest point. The balance lid (cover) is to be closed to prevent drafts from affecting the oscillations.

2. Place a 6mg weight on the right pan. The rest point should not shift less than one division on the indicator scale.
3. Level the balance and determine the rest point with a 10g. Weight in the centre of each pan.
4. Place a 6mg weight on the right pan. Again the rest point should not be shifted less than one division.
- For balances with sets of weights not less than 10 mg. The following practical procedure is suggested.
 1. Record the number of scale divisions the index pointer is shifted by a 30mg, a 20mg and a 10mg. Weights.
 2. Plot the weight against the number of scale divisions shifted by each weight.
 3. Draw a straight line through the points and from the straight line read the weight that will cause a shift of one scale division.
 4. Repeat with a 10g weight on each pan.

Volume

- It is possible to describe a given amount of matter by the quantity of space it occupies (its volume), as well as by its weight. But the custom is to measure amounts of liquids by volume and solids by weight.
- An important factor in the accuracy of an instrument used for measuring volume is the surface area of the liquid in it.
- Accuracy increases as surface area decreases. However, a decrease in surface area decreases the convenience of transferring a liquid to and from the equipment. Therefore, accuracy and convenience should be compromised
 - Volume is measured by using measures.
 - There are several types of measures. Some of them are the following:

Involves assuring the identity of ingredients by constant vigilance. It requires:

- Cylindrical Graduates (Graduated Cylinders),
- Conical Graduates

- Burettes
- Graduated
- Pipettes
- Droppers
- Volumetric Flasks
- Conical Flasks.

Techniques of measuring liquids

Good technique in measuring liquids involves the following points. The instructions given below are for right-handed personnel.

- Good technique in measuring liquids involves the following points. The instructions given below are for right-handed personnel.
 1. Choose the smallest measure that will hold the required volume. If possible do not split the volume between two measurements because this will increase the error.
 2. Hold the bottle with the right hand. Keep the label uppermost so that it is visible during measuring and so that any trickles after use do not obscure the label.
 3. Remove and hold the stopper of the bottle with the little finger of the left hand. If it is not possible to hold the stopper., it should be placed top uppermost on a clean tile to protect against atmospheric contamination
 4. Hold the graduate, which should be clean and dry, at the bottom with the thumb and the forefinger, and support it on the curved middle finger of the left hand. Don't tilt the measure.
 5. Raise the graduate so that the required mark is level led with the eyes to minimize errors in reading (Error or parallax).
 6. Pour the liquid into the center of the graduate until the bottom of the meniscus exactly reaches the required mark.
For very dark liquids, the true meniscus is often invisible and it is recommended that such liquids should be measured by difference.
 7. Close the stock container and return it tidily to the shelf.

Meniscus Volume:

The volume above the horizontal tangent touching the lowest point of the surface of a liquid is the meniscus volume.

EXPERIMENT VI**Interpretation of Prescriptions**

Prescription is an order signed by a *licensed medical practitioner*, directing the pharmacist to compound and dispenses a medicament.

A complete prescription includes the following:

1. Prescriber's office information
2. Date the prescription was written,
3. Name of the patient and address
4. Superscription: contains the traditional symbol Rx., meaning 'you take',
5. Inscription: Contains the list of ingredients and their quantities,
6. Transcription: contains the necessary directions to be labeled on the container for use by the patient
7. Name of the prescriber or his initials

Labeling

The label on dispensed medicine should provide the patient with all the information necessary to that the medicine may be taken or used appropriately.

Information on the label includes:

1. Patient's name
2. Name and address of pharmacy
3. Date of dispensing
4. Name, strength & quantity of the preparation
5. Directions for use
6. Storage condition and shelf life (expiry date)

7. Cautionary and advisory labels.

The information on the labels of dispensed medicines should be accurate, legible, intelligible, adequate and relevant.

1. Accurate: The label should be checked immediately after writing and again prior to fixing to container.
2. Legible: The label should be typewritten or printed or written neatly with pen.
3. Intelligible: The information should be completely unambiguous and arranged to avoid confusion. The terms should be readily understandable by the lay person.
4. Adequate and relevant: Care should be taken to avoid causing confusion and anxiety. Too much information on the label may mean that none of it is noted.
 - Note that the size of the label should be related to the size of the container.

EXPERIMENT VII

Formulation of liquid Dosage Forms

- Dosage forms are the means by which drug molecules are delivered to sites of action within the body.

Types of dosage forms

- Based on the physical nature of the products, dosage forms are classified into 3 groups.
 1. Liquid dosage forms: Solutions, Suspensions and Emulsions.
 2. Semi-solid dosage forms: Ointments, Pastes, Creams, Jellies.
 3. Solid dosage forms: Powders, Granules, Capsules, Tablets, Suppositories, etc.

Solutions

Solutions are homogenous liquid preparations containing one or more dissolved ingredients and are used for a variety of purposes as both internal and external dosage forms.

1. Solutions taken orally: mixtures, elixirs, syrups, linctuses, draughts
2. Solutions used in mouth and throat: mouth washes and gargles, throat paints, throat sprays.
3. Solutions instilled into body cavities: douches, enemas, eardrops, eye drops, nasal drops, nasal sprays.
4. Solutions applied to body surface: collodions, liniments, lotions, and paints.

- In addition to the active ingredients (medicaments), solutions also contain vehicles and adjuncts.
- The vehicle is the medium in which the ingredients of a medicine are dissolved or dispersed for solutions this is the solvent.
e.g. Water, Aromatic waters, Alcohol, Glycerol, Syrup, etc.
- Water is the vehicle of choice for the majority of pharmaceutical solutions. Because. Water is generally available, relatively inexpensive, palatable and non-toxic for oral use and non-irritant for external use. It also acts as a solvent for a wide range of substances.
- The adjuncts are additives that are added to improve the quality of the preparations.
e.g. Chemical stabilizers, Colorants, Flavorants, Preservatives, etc.

Dissolution Techniques

1. Check the Solubilities of the ingredients from an official book. Remember that some substances are less soluble at raised temperatures, Paraldehyde being an example.
2. Select a vessel in which to prepare the solution. A conical flask that will hold the final volume of the preparation is the preferred vessel. The advantages of conical flasks over other vessels are:
 - a. Contents can be agitated by shaking with little risk of spillage;
 - b. Contents can be readily heated if required;
 - c. Contents can be readily cooled under running water with little risk
of admitting water to the solution.
3. Powder the medicament (except for substances already in fine powder form or for substances known to be freely soluble). A porcelain mortar is generally preferred for size reduction, although glass mortars should be used for substances that taint or stain.

4. Weight the medicament. Note that powdering should precede weighing because of the difficulty in removing all the dry powder from the mortar. In order to avoid powdering a large excess of medicament, which would then be wasted, an approximate weight plus a little excess may be weighed and transferred to the mortar. For readily soluble substances to be dissolved in the mortar, powdering may follow weighing, e.g. Menthol in Eucalyptus oil.
5. Agitate the solution. Shaking the flask or stirring with a glass rod will aid solution. For slowly soluble substances a mechanical stirrer is to be preferred. A simple stainless steel propeller may be derived by a variable speed electric motor.
6. Heat the solution (only if necessary and if it does not damage the medicament). Heating is appropriate for substances that are:
 - a. more soluble at high temperature, e.g. proflavine;
 - b. non-heat labile;
 - c. non-volatile (unless special precaution are taken).
7. Check that solution is complete
 - Examine the solution critically for traces of undissolved solid on the neck of the flask, in the bottom of the flask or in suspension in the solvent.
 - Traces of powders on the inside shoulders or neck of the flask must be washed into the bulk of the liquid.
8. Restore the solution to room temperature (if necessary).
 - Solutions that have been heated must be cooled before adjustment to the final volume.
9. Transfer to a measure and adjust to final volume.
 - It is more accurate to adjust to volume in a measure. However, for very viscous preparations, e.g., linctuses, where transfer from a measure to the container might leave a significant volume on the inside of the measure (in spite of careful draining), adjustment may be done in the container.
 - The container is placed on a flat surface and tarred by carefully adding the correct volume of water and making the level of the true meniscus with a small

level. Unless the preparation contains water, the calibrated container must be drained well before use and, in some instances, drying may be necessary.

Special techniques for volatile substances

- **Volatile solvents:** Some solvents contain volatile ingredients (e.g., chloroform water and peppermint water) and should not be heated if this can be avoided. However, as most of these waters are made at the time of use by diluting a concentrated solution with water it may be practicable to dissolve the medicament in hot water and add the concentrate after cooling.
- **Volatile solutes:** A number of very volatile substances, of which chlorbutol and chlorcresol are commonly, used examples, present special problems. They dissolve very slowly in the cold even with size reduction and mechanical stirring. When solutions must be made quickly, heat is required. A suitable' small scale method is as follows:
 - a. Measure as much water as the formula allows and transfer to a conical flask suitable for stoppering.
 - b. Heat the water to approximately 60°C. The flask must not be heated with the stopper in place.
 - c. Add to the hot water the substance to be dissolved.
 - d. Quickly, insert the stopper and shake vigorously until solution is complete.

Aromatic Water

- Aromatic waters are aqueous solutions, usually saturated, of volatile oils or other volatile substances that are characterized by very low water solubilities. e.g. chloroform
- Aromatic waters are used externally as perfumes (e.g. Rose Water), and internally as flavoring agents (e.g. Peppermint Water, Cinnamon Water and Orange Flower Water) and/or as medicaments (e.g. Peppermint Water has been used as a carminative and Chloroform Water was used in expectorant preparations).

Methods of Preparation

The U.S.P. prescribes 3 general methods of preparation of aromatic waters: *Distillation*, *Solution and Alternate Solution*. A fourth method, *Dilution*, is also recommended by the B.P.C.

1. Distillation

- Place the odoriferous portion of the plant or drug in a suitable still with sufficient purified water.
- Distill most of the water, carefully avoiding the development of empyreumatic odors through charring or scorching of the substances.
- Separate the excess oil from the distillate.
- Then the aqueous phase is the product.

2. Solution

- Agitate the volatile substance with purified water for a period of 15 minutes
- The mixture is set aside for at least 12 hrs to ensure saturation
- Then filter through wetted filter paper. The filter paper must be wet to prevent the passage of excess oil into the filtrate and to eliminate absorption of dissolved aromatics by the filter.

3. Alternate Solution

- Mix thoroughly the volatile material with 15g of purified talc.

Then agitate the mixture with a liter of purified water for 10 min., prior to filtration

- The timesaving factor is an important advantage of this method.

4. Dilution

Preparation of Concentrated Aromatic Waters

- Mix an alcoholic solution of the volatile oil with purified water and add talc.
- Agitate the mixture and set aside for several hours

- Filter and the filtrate is call *concentrated aromatic water*. The concentrate contains 50 to 55 percent alcohol by volume.
- One volume of concentrated aromatic water is diluted with 39 volumes of water, producing the aromatic water, which contains less than 1.5% of alcohol.

Practical

1. (a) Chloroform Water, B.P. 88		
Chloroform	-----	2.5 ml
Water FBC to	-----	1000 ml
	Send 100ml.	

(b) Double Strength Chloroform Water, B.P.88		
Chloroform	-----	5 ml
Water FBC to	-----	1000 ml
	Send 100ml.	

Compounding: Dissolve the chloroform in purified water by shaking.

2. Peppermint Water		
Pepper mint oil		0.2 ml
Water, FBC to	-----	100 ml
	Send 50ml.	

Compounding: Dissolve the oil in purified water by shaking.

3. Concentrated Camphor Water, BP. 88		
Ingredients		Master Formula
Camphor	-----	40 g
Alcohol (90%)	-----	600 ml
Water, FBC to	-----	1000 ml
	Send 50ml.	

Compounding: Dissolve the camphor in the alcohol. Then, add water gradually in successive portions, vigorously shaking after each addition.

4. Camphor Water, NF.		
Ingredients		Master Formula
Camphor	-----	2 g
Water, FBC to	-----	1000 ml
	Send 100ml.	

Compounding: Incorporate suitably comminuted camphor with 15g of talc. Then, add the required volume of purified water and agitate vigorously several times during 10 minutes period and filter.

5. Concentrated Cinnamon Water, BP. 88

Ingredients		Master Formula
Cinnamon oil	-----	20 ml
Alcohol (90%)	-----	600 ml
Water, FBC to	-----	1000 ml

Compounding: Dissolve the oil in the alcohol. Add the water in successive portions, shaking vigorously after each addition. Then, add 50g of talc, shake occasional during few hours and finally filter.

EXPERIMENT VIII

Spirits, Tinctures and Mixtures

1. Spirits

- Spirits, also known as essences, are alcoholic or aqueous alcoholic solutions of volatile substances. Like the aromatic water, the active ingredients in the spirits may be a solid, liquid or gas. The volatile substances in the majority of cases are volatile oils.
- Some spirits are used internally for their medicinal value, a few medicinally by inhalation. But a large number are used as flavoring agents.
- The amount of volatile material in spirits varies greatly and no fixed percentage can be given. However, in all cases, the volatile oil content of the official spirits is much greater than that of the corresponding aromatic waters.

Preparation

Most spirits are prepared by simple solution in alcohol, although some can be prepared by distillation, maceration, and chemical reaction methods.

In preparing spirits, it must be kept in mind that the oils dissolved in alcohol, are precipitated, causing turbidity when the solutions are mixed with water. In order to avoid this turbidity, water, except as specified in the formula, should be avoided.

Graduates and other equipment used should be thoroughly dry. Filter paper should be moistened with alcohol.

Storage

Spirits should be kept in well-closed containers and stored at temperatures not exceeding 25°C. Good storage conditions prevent evaporation and volatilization of either the alcohol or the active principle(s).

Practical

Peppermint Spirit, B.P.88 (peppermint Essence)

Ingredients	Master Formula
Peppermint oil	100ml
Ethanol (90) to	1000ml
Send 30ml.	

Compounding: Dissolve the peppermint oil in ethanol. Add sufficient ethanol to produce the final volume. If the solution is not clear, shake with previously sterilized purified talc and filter.

Action and Uses: The spirit is an effective carminative. Also, mixed with other drugs or preparations, it may be used as a flavor.

2. Benzaldehyde Sprit

Ingredient	Master Formula
Benzaldehyde	10 ml
Ethanol (90%)	800ml
Purified water to	1000ml
Send 50ml.	

Compounding: Dissolve the benzaldehyde in the ethanol and add sufficient purified water to produce the final volume.

3. Chloroform Spirit

Ingredient	Master Formula
Chloroform	50ml.
Ethanol (90%)	1000ml.
Send 40ml.	

Compounding: Mix the chloroform with the ethanol.

Uses and actions: flavoring agent and preservative.

2. Tinctures

- Tinctures are alcoholic or hydroalcoholic solutions of principles extracted from natural sources or of pure chemicals merely dissolved in these solvent systems, an example of the latter being Iodine Tincture.
- Tinctures vary widely in their method of preparation, the strength of their volatile ingredients, their alcoholic content and their intended use.
- Tinctures are prepared by a number of processes: Maceration (Process M), Percolation (process P) Solution, and Dilution.

Percolation (Process P)

Carefully mix the ground drug or mixture of drugs with sufficient quantity of the prescribed solvent or solvent mixture to render it evenly and distinctly damp, allow it to stand for 15 minutes, transfer it to a suitable percolator, and pack the drug firmly, Pour on enough of the prescribed solvent mixture to saturate the drug, cover the top of the percolator and, when the liquid is about to drip from the percolator, close the lower orifice, and allow the drug to macerate for 24 hrs or for the time specified in the monograph. Then allow the percolation to proceed slowly gradually adding sufficient solvent or solvent mixture to produce 1000ml of tincture, and mix.

Maceration (Process M)

Macerate the drug with 750ml of the prescribed solvent or solvent mixture in a container that can be close, and put in a warm place. Agitate it frequently during 3 days or until the soluble matter is dissolved. Transfer the mixture to a filter, and when most of the liquid has drained away, wash the residue on the filter with a sufficient quantity of the prescribed solvent or solvent mixture, combining the filtrates, to produce 1000ml of tincture and mix.

- Tinctures should be stored in tight, light resistant containers protected from direct sunlight and Excessive heat.

Practical

1. Iodine

Ingredients	Master Formula
Iodine	25g
Potassium iodide	25g
Purified water	25mg
Ethanol (90%) to	1000ml
Send 50ml.	

Compounding: Dissolve the potassium iodide and the iodine in the purified water and add sufficient ethanol (90 %) to produce 1000ml.

Actions and uses: Antiseptic.

2. Orange Tincture, BP,88

Ingredients	Master Formula
Dried Bitter-Orange Peel, in moderately fine powder	110g
Ethanol (70%) to	1000ml

Compounding: Percolation method.

Use: Flavoring agent.

3. Mixtures

- Mixtures are liquid oral preparation containing one or more active ingredients dissolved and /or suspended in water or some other aqueous vehicles such as aromatic waters.
- Frequently, the term mixture is applied loosely to aqueous preparations of description.

The U.S.P. does not recognize the term, mixture. The term, suspension now is used to describe a number of similar preparations.

- Official mixtures are not usually formulated for a long shelf life and should be recently prepared.

Practical

1. Sodium Salicylate Mixture, BP, 88 (Sodium Salicylate Oral Solution)

Ingredients	Master formula
Sodium Salicylate	50g
Sodium Metabisulfite	1g
Concentrated Orange Peel Infusion	50ml
Double Strength chloroform water	500ml
Water FBC, sufficient to produce	1000ml
M.Ft. M., Mitt. 50ml	

Formulation: sodium salicylate mixture is a solution containing 5% W/V of sodium salicylate and a suitable antioxidant in a suitable vehicle with an orange flavor.

Compounding: Dissolve the sodium salicylate and sodium metabisulfite in the water. Add the concentrated orange peel infusion and the double strength chloroform water. Finally adjust to the desired volume.

Actions and Uses: Anti-inflammatory and Analgesic.

2. Magnesium Sulfate Mixture, BP.88 (Magnesium Sulfate Oral Suspension)

Ingredients	Master Formula
Magnesium sulfate	400g

Light magnesium carbonate	50g
Concentrated Peppermint Emulsion	25ml
Double strength Chloroform Water	300ml
Water FBC, sufficient to produce	1000ml
M.Ft.M, Mitt. 50ml.	

Sig. 10 ml t.i.d., shake the bottle before use

Formulation: Magnesium sulfate mixture is a suspension containing 40%w/v magnesium sulfate and 5%w/v of light management carbonate in a suitable vehicle with peppermint flavor.

Compounding: Mix the magnesium sulfate and light magnesium carbonate in a mortar and pestle and triturate the mixture with some water. Transfer the mixed mixture to a measure and double strength chloroform water and add concentrated peppermint emulsion. Finally adjust to the final volume with the water.

Actions and Uses: A laxative.

EXPERIMENT IX

Syrups, Elixirs and Linctuses

1. Syrups

- Syrups are concentrated aqueous solutions of sucrose, other sugars or sweetening agents, to which small quantities of suitable polyhydric alcohols, may be added to retard crystallization or to increase the solubility of the other ingredients.
- Syrups usually contain aromatic or other flavoring materials
- Medicinally, syrups are divided into two groups: the *flavoring* syrups, which are used as vehicles, and the *medicated* syrups, which contain ingredients giving them therapeutic value.
- Pharmaceutically, syrups are classified best according to their basic formulas: those which are concentrated solutions of a sugar (Sucrose, Dextrose) and those which are formulated with artificial sweetening agents and viscosity builders.
- Generally, syrups are stored at room temperature (not exceeding 30°C) in tightly stoppered well-filled bottles.
- Nearly saturated solutions of sucrose, if stored properly, are self-preserving. But, in dilute solution, sucrose, as a carbohydrate, provides a nutrient medium for the growth of microorganisms, particularly yeasts and molds. Therefore, preservatives should be added if the concentration of sucrose is significantly low.

Practical

(1) Simple Syrup, U.S.P.	
Sucrose -----647.4 g	Sucrose -----850 g
Purified Water to -----1000 g	Purified Water to --1000 ml

Formulation:

- Simple Syrup, U.S.P. contains 85% w/v of sucrose, corresponding to 64.74% w/w. Its specific gravity is 1.313. The latter parameter provides a ready means of standardizing the preparation.
- A saturated solution of sucrose is 67.9% w/w at 25°C; the British Pharmacopoeia Syrup is 66.7% w/w sucrose.
- Sucrose is subject to two degradative pathways in aqueous solutions: Fermentation and Hydrolysis.
- In dilute aqueous solution, sucrose, as a carbohydrate, provides a nutrient medium for the growth of microorganisms, particularly yeasts and molds. The consequences of this growth include turbidity, fermentation, and changes in taste.
- Concentration of sucrose is important, i.e., nearly saturated solutions of sucrose, if stored properly, are self-preserving. In effect, such solutions contain no ‘free water’; thus, they behave as an anhydrous medium with respect to microorganisms.
- Preservatives may be specified in syrup formulations to prevent fermentation; however, most, official syrups depend only on sufficient concentration of sucrose for preservation. If the concentration of sucrose is significantly less than that of Syrup, U.S.P., preservatives should be added.
- **N.B.** The more free water, the more preservative required in the product. In some syrups, alcohol is present in small amounts. It serves as a solvent for the alcohol-soluble ingredients. Although the concentration of alcohol is not sufficient to

have a preservative effect, alcohol concentrates in the vapors above the syrup and thus prevents the growth of surface molds.

- Simple Syrup is a saturated solution at 4°C. Therefore, no crystallization should be observed in properly prepared syrup, unless the temperature drops below 4°C. If the syrup is too concentrated, crystallization may be observed; if too dilute, Simple Syrup will readily support the growth of molds.

Methods of Preparation

- A. **Hot Process:** Place the sucrose in a dish, add the water and dissolve the sucrose by heating on a water bath, stirring continuously. When dissolution of the sucrose is complete, allow the solution formed to cool and add sufficient boiling purified water to make up for that lost by evaporation. Then strain the syrup, while still warm, through a fine muslin supported in a funnel into a bottle which is capable of nearly filling, and shake the bottle occasionally until the content is quite cold.
- B. **Cold Process:** Place the sucrose in a suitable percolator the neck of which is nearly filled with loosely packed cotton moistened, after packing, with a few drops of water. Pour carefully about 450 ml of purified water upon the sucrose, and regulate the outflow to a steady drip of percolate. Return the percolate, if necessary, until all the sucrose has dissolved. Then wash the inside and the cotton with sufficient purified water to bring the volume of the percolate to 1000 ml and mix.
- C. **Alternatively,** the sucrose may be dissolved more rapidly in the proper amount of water by agitation in a graduated bottle.

N.B. Syrup made without heat is practically colorless while syrups made with heat have pale amber color due to the levulose formed as a result of hydrolysis of sucrose.

(2)

Chloral Syrup, B.P.C. 68	
Chloral hydrate	200 g

Water, FBC	200 ml
Syrup to	1000 ml
Ft. Syrup, Send 20 ml	
Sig. Two 5 ml h.s.	

Compounding: Dissolve the chloral hydrate in the water, and add syrup to volume.

Note: Chloral hydrate is volatile and sensitive to light, so the syrup must be recently prepared.

Actions and Uses: Hypnotic.

(3)

Ammonium Chloride Cough Syrup	
Ammonium chloride	4.0g
Liquid Extract of Liquorice	5.0 ml
Syrup	50.0 ml
Water q.s. to	100.0 ml
Ft. Cough Syrup 25 ml	
Sig. 1tsp. t.i.d.	

Compounding: Dissolve the ammonium chloride in portion of the water, and add the liquid extract of liquorices, the syrup and water to adjust to volume.

Actions and Uses: Expectorant.

(4) Ferrous Sulfate Syrup, U.S.P.

Ingredients	Master Formula
Ferrous sulfate	40 g.
Citric acid, hydrous	2.1 g
Peppermint spirit	2 ml
Sucrose	825 g
Purified water to	1000 ml

Ft. Syrup, Sende 50 ml	
Sig. Two 5 ml t.i.d.	

Formulation: The peppermint spirit, used as the source of flavor, has a persisting effect, which masks the somewhat unpleasant ferruginous taste of the Fe^{2+} . Care must be taken in preparing aqueous solutions of ferrous salts, since Fe^{2+} is readily oxidized by dissolved oxygen to form Fe^{3+} which precipitates as basic ferric salts; discoloring the solution. Ferrous sulphate is dissolved in purified water containing some sucrose, which provides a reducing environment thereby inhibiting auto-oxidation. If all of the sucrose were added, the solution would be too viscous to allow perfect filtration. Citric acid is included in the formulation to prevent discoloration of the syrup i.e. a change from a green to a reddish brown tint; it does this by preferentially chelating Fe^{3+} .

Compounding: Dissolve the ferrous sulfate, the citric acid, the peppermint spirit, and 200g of the sucrose in 450ml of purified water, and filter the solution until clear. Dissolve the remainder of the sucrose in the clear filtrate, and add purified water to make 1000ml. Mix, and filter, if necessary through a pled get of cotton.

Actions and Uses: Haematinic

2. Elixirs

- Elixirs are clear, flavored oral liquids containing one or more active ingredients dissolved in a vehicle that usually contains a high proportion of sucrose or a suitable polyhydric alcohol or alcohols (glycerol, Propylene glycol) and may also contain ethanol.
- Elixirs generally owe their pleasant flavor to the presence of sugar and the volatile flavoring agents. However, the sugar content of elixirs is lower than that of syrups.
- In general elixirs are reasonably stable preparations provided they are stored in well-filled containers and are not diluted or mixed with other preparations.

Preparation of Elixirs

Elixirs are usually prepared by simple solution with agitation and/or by the admixture of two or more liquid ingredients. Alcohol-soluble and water-soluble components are

generally dissolved separately in alcohol and purified water respectively. Then the aqueous solution is added to the alcoholic solution, rather than the reverse in order to maintain the highest possible alcoholic strength at all times so that minimal separation of the alcohol-soluble components occurs. When the two solutions are completely mixed the mixture is made to volume with the specified solvent or vehicle.

Practical

(1) Phenobarbitone Elixir

Ingredients	Master Formula
Phenobarbitone	0.4 g
Ethanol (90%)	40 ml
Compound orange spirit	2.5 ml
Glycerol	40 ml
Amaranth solution	1 ml
Purified water, to	100 ml
Ft. elixir, Send 25 ml	
Sig. 2tsp. h.s., store in dark place	

Formulation: Ethanol (90%) and glycerol are included as co-solvents for the phenobarbitone. Compound orange spirit is used with the glycerol for flavor and the amaranth solution for color.

Phenobarbitone is virtually insoluble in water, but a clear product can be made by dissolving it in alcohol and then diluting with glycerol and water.

Compounding: Dissolve the phenobarbitone in the ethanol (90%) and add the compound orange spirit before adding the glycerol, the amaranth solution and water to volume.

Actions and Uses: Anti-convulsant.

Note: Phenobarbitone causes drowsiness; alcohol should be avoided.

2. Ephedrine Elixir, B.P.C.

Ephedrine hydrochloride	3.0 g
Glycerin	200.0 ml
Invert syrup	200.0 ml
Alcohol (90%)	00.0 ml
Purified water	60.0 ml
Chloroform Spirit	40.0 ml
Amaranth solution	10.0 ml
Lemon Spirit	0.2 ml
Syrup to	1000.0 ml

Compounding: Dissolve the ephedrine hydrochloride in water, add the glycerin, the Amaranth solution, the alcohol, the chloroform spirit, the lemon spirit, the invert syrup and sufficient syrup to produce the required volume, and mix.

3. Linctuses

- Linctuses are viscous oral liquids that may contain one or more active ingredients in solution. The vehicle usually contains large amounts of sucrose, other sugars or a suitable polyhydric alcohol or alcohols.

- Linctuses are intended for use in the treatment or relief of cough, being sipped and swallowed slowly without the addition of water.

Practical

1. Simple Linctus, B.P.	
Concentrated Anise water	10 ml
Amaranth solution 1%	15 ml
Citric acid monohydrate	25 g
Chloroform spirit	60 ml
Syrup to	1000 ml
Sende 40 ml	

Formulation: The concentrated anise water, the citric acid and the chloroform water act as flavoring agents, the amaranth solution as coloring and the syrup as a demulcent vehicle.

Compounding: The citric acid should be dissolved in the chloroform spirit, conc. Anise water and amaranth solution before adding the syrup to volume.

Action and Uses: Demulcent in the treatment of cough.

To be sipped and swallowed slowly undiluted.

EXPERIMENT X

Solutions Used in the Mouth: Gargles and Mouthwashes

1. Gargles

- Gargles are aqueous solutions intended for use in the prevention and/or treatment of throat infection (by forcing air from the lungs through the gargle, which is held in the throat).
- Gargles usually contain a bactericide, e.g. phenol or thymol, and are dispensed in concentrated forms with directions for dilution with warm water before use.

Practical

1. Phenol Gargle (Carbolic Acid Gargle)

Ingredients	Master formula
Phenol Glycerin	50 ml
Amaranth Solution	10 ml
Water to	1000 ml
Mitt. 50ml	
Sig. Dilute with an equal volume of warm water before use.	
Not to be swallowed in large amounts.	

Compounding: Mix the phenol glycerin and the amaranth solution with portion of the water. Agitate and add water to final volume.

Actions and Uses: Antibacterial and anesthetic.

1. Potassium Chlorate and Phenol Gargle

Ingredients	Master Formula
Potassium Chlorate	30 g
Patent Blue V	0.01g
Liquefied phenol	15ml
Water to	1000 ml
Ft. garg. Mitt. 50 ml. Sig. Dilute with ten times its volume of warm water before use. Not to be swallowed in large amounts.	

Compounding: Dissolve the potassium chlorate in about 35ml of warm water. Cool before adding the liquefied phenol. Add the dye and make up to 50ml with water.

Note: The most convenient way of obtaining the tiny amount of the Patent Blue V required for this preparation is to use a stock solution. Phenol is caustic so don't use mouth for pipe ting.

Actions and Uses: Mild astringent and antibacterial.

2. Mouth Washes

- Mouthwashes are aqueous preparations used to cleanse and deodorize the buccal cavity for the purpose of oral hygiene and to rect infections of the mouth.
- Mouthwashes are particularly refreshing to bed-ridden patients; like gargles, they are usually used after dilution with warm water and directions for diluting the mouthwashes should be given on the labels of these preparations.

1. Compound Sodium Chloride Mouthwash

Ingredients	Master Formula
-------------	----------------

Sodium Bicarbonate	10 g
Sodium Chloride	15g
Concentrated Peppermint Emulsion	25ml
Double Strength Chloroform Water	500 ml
Water to	1000 ml
Mitt. 50 ml. Sig. Dilute with an equal volume of warm water before use.	

Compounding: Dissolve the sodium bicarbonate and the sodium chloride in portion of the water. Add the conc. Peppermint emulsion and the double strength chloroform water. Finally adjust to final volume with water.

Actions and Uses: Cleansing and deodorizing.

2. Alkaline Phenol Mouthwash

Ingredients	Master Formula
Liquefied Phenol	30 ml
Potassium Hydroxide Solution 5%	30 ml
Amaranth Solution	1ml
Water to	1000 ml
Mitt. 50 ml. Sig. Dilute with ten times its volume of warm water before use. Not to be swallowed in large amounts.	

Compounding: Mix the liquefied phenol, the potassium hydroxide solution, and the amaranth solution. Then add water in successive portions while agitating.

3. Zinc Sulfate and Zinc Chloride Mouthwash

Ingredients	Master
-------------	--------

	Formula
Zinc Sulfate	20 g
Zinc Chloride	20 g
Compound Tartrazine Solution	10ml
Dilute Hydrochloric Acid	10ml
Chloroform Water, DS	500 ml
Water to	1000 ml
Send 50 ml. Sig. Dilute with twenty times its volume of warm water before use.	

Compounding: Dissolve the zinc sulfate and zinc chloride in portion of the water. Add the dilute hydrochloric acid, the compound tartrazine solution, the chloroform water, DS and adjust to final volume with water.

Note: Zinc chloride is very deliquescent and it is advantageous to keep the container in a desiccators. Crush a stick quickly in a mortar, weigh the required amount on a counterbalanced water glass and immediately return the excess to the bottle, which should be tightly closed. Zinc chloride usually contains some oxychloride, which makes the solution turbid; this disappears when the dilute hydrochloric acid is added.

Actions and Uses: Astringent.

EXPERIMENT XI

Solutions Instilled into Body Cavities: Douches, Enemas, Ear Drops and Nasal drops

1. Douches

- A douche is an aqueous solution used for rinsing a body cavity and functions as cleaning agent, antiseptic, or astringement.
- The word *douche* is most often used for vaginal solutions but it is sometimes applied to solutions for the bladder (usually called irrigations) or the rectum (usually called enemas).
- Douches usually are directed to the appropriate body part by using bulb syringes.
- Douches are often supplied as *liquid concentrates* or *powders* to be diluted or dissolved, in the appropriate amount of warm water, to the correct strength and suitable volume (usually 1-2 liters) prior to use. However, tablets for preparing solutions are also available. If powders or tablets are supplied, they must be free from insoluble material, in order to produce a clear solution. Tablets are produced by the usual processes, but any lubricants or diluents used must be readily soluble in water. Boric acid may be used as a lubricant and sodium chloride normally is used as a diluent. Tablets deteriorate on exposure to moist air and should be stored in airtight containers.

1. Potassium Permanganate Vaginal Douche

Ingredients	Master Formula
Potassium Permanganates	1g
Water, F.B.C. to	1000ml
Mitt. 250ml	
Sig. Dilute with three times its volume of warm water Discard the remain after one week (7 days) Store in cool place. Not to be taken	

Note: Potassium permanganate solution will stain plastic surfaces; take appropriate precautions. Care must be taken to avoid spillage during weighing, and apparatus should be kept on a large glass or porcelain tile during preparation of the solution.

The equipment and container must be spotlessly clean because this solution will react with oxidizable residues and turn brown with loss of strength.

The closure of the container must resist oxidation. Use a screw cap containing a thick polythene wad. The old-fashioned glass stoppered bottle is also suitable. Use a permanganate-resistant wad in the cap.

Compounding: The solute dissolves slowly in water and since it may not be heated, the use of an electric stirrer is recommended. As an alternative a glass mortar may be used:

Transfer the medicament to a glass mortar and grind the crystals with water.

Allow undissolved crystals to settle and pour supernatant to a flask. Add more water to the mortar, regrind and decant, repeating these procedures until all the solid has dissolved and the mortar is free from color. Finally, filter the solution through a clean sintered glass filter and make up to volume through the filter.

Action and uses: Mild antiseptic and deodorant.

Advice for patients when dispensed: The solution stains skin, hair, fabric, etc.

2. Alum Douche

Ingredients	Master Formula
Alum	4g
Zinc Sulfate	4g
Liquefied Phenol	5ml
Glycerin	125ml
Water to	1000ml
Mitt. 100ml.	

Compounding: Dissolve the alum and the zinc sulfate in portion of the water. Add the liquefied phenol, glycerol and finally adjust to volume with water.

Actions and uses: Mild astringent and antiseptic.

2. Enemas

- Enemas are aqueous or oily solutions, suspensions or emulsions of medicaments intended for rectal administration for therapeutic, cleansing or diagnostic purposes.
- The cleansing preparations are used to evacuate feces in case of constipation or before an operation (Evacuation enemas). Cleaning enemas act in one of two ways.
- By stimulating peristalsis because either

1. Their volume is large (0.5 to 1 liter) or
2. They cause osmotic retention of water in the bowel. These types are of smaller volume, e.g., sodium phosphates enema (100ml) and magnesium sulfate enema.

- By lubricating impacted feces e.g. olive and arachis oil enemas

(100 to 550ml).

- The therapeutic preparations are used to influence the general system by absorption or Retention enemas or affect locally the seat of disease. Retention enemas, since they are to be retained in the intestine, they should not be used in larger quantities than 150ml for an adult. Usually, the volume is considerably smaller such as a few milliliters. Microenema is a term used to describe these small-volume preparations. Therapeutic enemas may be used as sedatives, antihelmintics, anti-inflammatory agents, nutrients, etc.
- Enemas used for diagnostic purposes contain radiopaque substances for x-ray examination of the lower bowel, e.g., Barium Sulfate Enema.
- Enemas are administered by means of enema syringes. Large volume enemas should be warmed to body temperature before use, to avoid chills and burns, and are injected slowly

2. Sodium Phosphates Enema

Ingredients	Master Formula
Sodium Acid Phosphate	160g
Sodium Phosphate	60g
Purified Water, FBC to	1000ml
Mitt. 50ml.	

Compounding: Dissolve the salts in the water.

Actions and uses: Osmotic laxative.

3. Paraldehyde Enema

Ingredients	Master Formula
Paraldehyde	10 ml
Sodium Chloride	0.81g
Water, to	100ml
Mitt. 50ml. Sig. Store in cool, dark place. Discard the remain after 2 days. Not to be taken, for rectal use only	

Note: Paraldehyde is more soluble in cold water than in warm water. So it is helpful to cool the vehicle. If a solute gives off heat during the process of solution (an exothermic reaction), its solubility is decreased with an increase in temperature.

- Because of the instability of paraldehyde, they enema should be stored in a cool place in complete darkness and an expiry date of only two days after issue is advisable.
- Because the volume is small, it is not essential to warm this solution above room temperature before administration; nor is this advisable because some of the paraldehyde may separate due to its lower solubility at raised temperatures.
- Paraldehyde oxidizes to acetic acid on storage and deaths have resulted from using old stock. It must be discarded if it becomes discolored or develops a definite odor of acetic acid.

- Paraldehyde has a very disagreeable taste and is irritant to the alimentary tract.

Compounding: Weigh the sodium chloride and dissolve it in a suitable volume of water that has been chilled in a refrigerator, preferably overnight, to aid solution of paraldehyde (Leave the water in the refrigerator until the salt has been weighed).

Put the paraldehyde in a bottle, adjust to volume with the vehicle and shake vigorously until solution is complete.

N.B. Chilling and immediate use of the maximum volume of vehicle are helpful because the concentration of paraldehyde is near to the limit of its solubility. Paraldehyde attacks cork and certain of the plastics used as liners for screw caps; wads of polyethylene (alone) are satisfactory.

Actions and uses: Anticonvulsant.

3. Nasal Drops, Ear drops and Nasal Sprays.

- Eardrops are solutions or suspensions of medicaments for instillation into the ear with a dropper. However, most eardrops are simple solutions. They contain medicaments for treating mild infections (e.g. chloramphenicol), softening wax (e.g., hydrogen peroxide or sodium bicarbonate), cleansing after infections (e.g., spirits), drying weeping surface (e.g.) the astringent, aluminum acetate), antiseptics and anesthesia (e.g., phenol).
- The Vehicle is usually water but glycerin may be included for its softening effect on the cerumen wax and its high viscosity, which assists adherence to infected surfaces. Chloramphenicol is soluble 1 in 400 in water and 1 in 7 in propylene glycol; to obtain a solution of effective strength, propylene glycol is used as the solvent.

- Nasal drops are liquid preparations for instillation into the nostrils by means of a dropper. Nasal drops usually contain substances with antiseptic, local analgesic, or vasoconstrictor properties.

They are usually aqueous because oily drops inhibit movement of cilia in the nasal mucosa and, if used for long periods, may reach the lungs and cause lipoidal pneumonia.

- Iso-osmotic with 0.9% sodium chloride
 - On the acid side of neutral (slightly buffered to maintain pH 5.5 to 6.5)
 - Similar viscosity to nasal mucus (0.5% of MC is sometimes used to thicken aqueous nasal drops).
- Nasal sprays are preparations of medicaments applied to the nose by means of atomizer. Their chief uses are to
 - Reduce nasal congestion and
 - To treat infection
 - The aim is to retain the solution in the nasal tract (contrast to throat sprays, which are aimed to treat the lungs). Therefore, a coarse spray produced by a scent-spray type of atomizer or a plastic squeeze bottle is most suitable.
 - As with nasal drops and throat sprays, oily preparations should be avoided in nasal sprays.
 - There are no nasal sprays in the official formularies.

1. Phenol Eardrops

Ingredients	Master Formula
Phenol Glycerin	40 ml
Glycerol, to	100ml
Mitt. 10ml. Sig. Store in a cool place.	

Not to be diluted with water, Not to be taken

Compounding: This is a viscous preparation. Either

1. Make an excess; the nearest volume that is accurately measurable in a conical flask is 15ml. or
2. Calibrate a bottle at 10ml, add the phenol glycerin by difference, make to volume with the glycerol, and mix. This method is less wasteful but has several disadvantages.
 - i. The bottle must be very carefully dried after calibration, because these drops become caustic if diluted with water and, if an oven is used for drying, a paper calibration label may fall off.
 - ii. Mixing with a rod through the narrow neck of a bottle is not very efficient and solution will be carried away on the rod. Shaking is unreliable if the closure carries a pipette type dropper because liquid becomes trapped in the pipette.

Container: Use a colored, fluted glass bottle (ribbed oval glass bottle) with a dropper in the cap.

3. Sodium Bicarbonate Eardrops

Ingredients	Master Formula
Sodium Bicarbonate	5g
Glycerol	30ml
Purified Water, FBC to	100ml
Mitt.2.	

Compounding: Dissolve the sodium bicarbonate in about 60ml of water. Add the glycerol and sufficient water to produce 100ml and mix.

Actions and uses: Soften cerumen wax.

4. Ephedrine Nasal Drops

Ingredients	Master Formula
Ephedrine Hydrochloride	0.5g
Chlorbutol	0.5g
Sodium Chloride	0.5g
Water, to	100ml
Mitt. 20ml Sig. Store in a cool place . For use in the nose only Not to be taken	

Compounding: Heat as much water as the formula allows to 60°C. add the chlorbutol, insert the stopper and shake vigorously until solution is complete. Then dissolve the salts in the warm water. Cool, filter if necessary, and make to volume through the filter.

Actions and uses: Reduce swelling of the nasal mucosa and underlying tissues in hay fever and catarrh.

EXPERIMENT XII

Solutions for External Use: Collodions, Liniments, Lotions, Paints

1. Collodions

- Collodions are liquid preparations containing pyroxylin (a nitrocellulose) dissolved in a solvent mixture of ethyl ether and ethanol with or without added medicinal substances,
- Collodions, when applied to the skin, the solvent rapidly evaporates, having a filmy residence of pyroxylin. This provides an occlusive protective coating to the skin, and when the collodion is medicated, it leaves a thin layer of that medication firmly placed against the skin.
- Collodions are applied to the skin by means of a soft brush or other suitable applicator and must be applied to dry tissues to affect adhesion to the skin's surface.
- Collodions must be clearly labeled "For External Use Only" or with words of similar effect.
- Storage: collodions should be kept in well-closed containers, at temperatures not exceeding 30°C and remote from fire.

1. Collodion

Ingredients	Master Formula
-------------	----------------

Pyroxylin	40g
Ether	750 ml
Alcohol	250 ml
To make about	1000ml
Send 100ml Sig. Store in a cool place. High flammable, keep away from naked flames	

Note: Colodion is a clear or slightly opalescent, viscous liquid. The product is capable of forming a protective film on application to the skin and the volatilization of the solvent. The film is useful in holding the edges of an incised wound together. However, its presence on the skin is uncomfortable due to its inflexible nature. The following product, which is flexible, has a greater appeal when a nonviable film is not required.

Compounding: Add the mixture of alcohol and ether to the pyroxylin contained in a suitable container, and insert the stopper into the container. Shake occasionally until the pyroxylin is dissolved.

Actions and uses: Skin protective.

2. Flexible Collodion

Ingredients	Master Formula
Camphor	20 g
Castor Oil	30 g
Collodion, to	1000g
To make about	1000ml
Mitt. 10g	

Sig. For external use only. Store in a cool place, Highly flammable, keep away from naked flames

Note: Flexible collodion is prepared by adding 2% of camphor and 3% of castor oil. The castor oil renders the product flexible, permitting its comfortable use over normally moved areas, such as fingers or toes.

Compounding: Weigh the ingredients, successively into a dry, tarred bottle. Shake the mixture until the camphor is dissolved.

Actions and uses: Skin protective

3. Salicylic Acid Collodion

Ingredients	Master Formula
Salicylic Acid	120 g
Flexible Colodion to	1000ml
Mitt. 10ml	
Sig. For external use only. Store in a cool place, Highly flammable, keep away from naked flames	

Note: Salicylic acid collodion is a 10% solution of salicylic acid in flexible collodion. It has keratolytic effects, especially in the removal of corns from the toes. The preparation is applied one drop at a time onto the corn or wart, allowing time to dry before another drop is added. Because salicylic acid can be irritating to normal, healthy skin attention should be given to ensure application directly onto the corn and wart. A useful preventive method is to cover adjacent healthy skin with some white petrolatum prior to application of the product. Proper tightening and storage of product after use is an absolute necessity because of the vehicle.

Compounding: Dissolve the salicylic acid in the flexible collodion.

Actions and uses: Keratolytic in the treatment of corns and warts.

1. Camphor Liniment

Ingredients	Master Formula
Camphor	200 g
Arachis Oil	800 g
Mitt. 10g. Sig. For external use only. Store in a cool place.	

Compounding: Dissolve the camphor in the oil.

Actions and uses: Counterirritant.

2. Ammoniated Camphor Liniment

Ingredients	Master Formula
Camphor	125 g
Strong Ammonia Solution	300 ml
Lavender Oil	5 ml
Alcohol (95%) to	1000 ml
Mitt. 50ml. Sig. For external use only. Store in a cool place.	

Compounding: Dissolve the camphor and the lavender oil in 500ml of the alcohol, add the strong ammonia solution gradually with frequent shaking, and then sufficient of the alcohol to produce the required volume.

Actions and uses: Counterirritant.

3. Methyl Salicylate Liniment

Ingredients	Master Formula
Camphor	125 g
Strong Ammonia Solution	300 ml
Lavender Oil	5 ml
Alcohol (95%) to	1000 ml
Mitt. 50ml. Sig. For external use only Store in a cool place.	

3. Lotions

- Lotions are liquid preparations intended for external application to the skin without friction (gently) by dabbing or after spreading on a dressing.
- Most lotions contain finely powdered through the use of suspending and dispensing agents. Other lotions have as the dispensed phase liquid substances that are immiscible with the vehicle and are usually dispersed by means of emulsifying agents or other suitable stabilizers.
- Lotions should be shaken vigorously before each use to redistribute any separated mater. Containers of lotions should be labeled to instruct the patient to shake thoroughly before use and also to use externally only.

1. Salicylic Acid Lotion

Ingredients	Master Formula
Salicylic Acid	20g
Castor Oil	10 ml
Alcohol (95%) to	1000 ml
Mitt. 50ml. Sig. For external use only	

Compounding: Dissolve the salicylic acid in portion of the ethanol; add the castor oil and sufficient ethanol to produce 1000ml and mix.

Actions and uses: Keratolytic for the treatment of dandruff.

2. Copper and Zinc Sulfates Lotion

Ingredients	Master Formula
Cooper Sulfate	15g
Zinc Sulfate	10 ml
Camphor Water to	1000 ml
Mitt. 50ml. Sig. For external use only	

Compounding: Dissolve the copper sulfate and the zinc sulfate in the camphor water.

Actions and uses: Astringents in the treatment of impetigo.

4. Paints

- Paints are solutions or dispersion of one or more active ingredients for application to the skin or, in some cases, mucous membranes. However, all the official paints are solutions.
- Skin paints often have a volatile solvent that evaporates quickly to have a dry or resinous film of the medicament.
- All paints, except throat paints, are labeled “For external use only”.

1. Brilliant Green & Crystal Violet Paint

Ingredients	Master Formula
Brilliant Green	5g
Crystal Violet	5g
Alcohol (90%)	500 ml
Camphor Water to	1000 ml
Mitt. 50ml. Sig. For external use only	

Note: Precautions to prevent spillage during dispensing are particularly necessary when a preparation contains a dye. Dyes are often light, fine powders that easily float into the air if carelessly handled. The spatula used for weighing should not be wiped withy cloth by but should be washed before putting down.

Compounding: Dissolve the brilliant green and the crystal violet in the alcohol and add sufficient water to produce the required volume.

Actions and uses: Antiseptic to sterilize the skin in gynecology.

2. Crystal Violet Paint

Ingredients	Master Formula
Crystal Violet	5g
Water to	1000 ml
Mitt. 50ml. Sig. For external use only. Apply undiluted	

Compounding: Dissolve the crystal violet in the water to produce the required volume.

Actions and uses: Antiseptic in the treatment of wounds and burns.

EXPERIMENT XV

Emulsions

- An emulsion is a disperse system consisting of two immiscible liquids, usually water and oil, one of which (the disperse phase) is distributed throughout the other (the continuous phase) as small globules.
- Since this type of dispersion is inherently unstable, the system is stabilized by the presence a third component, the emulsifying agent (emulgent, emulsifier).

Types of Emulsions

Emulsions can be classified into:

1. **Oil in water (O/W)**, in which the oil is dispersed in the water, i.e. the continuous phase is aqueous).
2. **Water in oil (W/O)** in which the water is dispersed in the oil, i.e. the continuous phase is oily.
3. **Multiple emulsion**, which could be of W/O/W or O/W/O
 - **Micro-emulsion:** If the dispersed globules are of colloidal dimension (1nm-1 μ m diameter), the preparation which is quite often transparent is called a micro-emulsion.

Determination of Emulsion Type

Tests for distinguishing between O/W and W/O emulsions include:

1. **Dilution Test (Miscibility Test):** It involves dilution of an emulsion with water or oil.
2. **Dye-solubility Test (Staining Test):** It involves the use of dyes soluble only in either the aqueous or the oil phase of an emulsion, but not in both phases, e.g. Scarlet red, Sudan III.
3. **Conductivity Test:** It uses the fact that an O/W emulsion possesses a much higher conductivity than a W/O emulsion.

Stability of Emulsions

A stable emulsion is one in which the globules

- Retain their initial character (i.e., mean size and size distribution); and
- Remain uniformly distributed throughout the continuous phase.

Instability in emulsion includes:

1. **Cracking (Breaking)** coalescence of the dispersed globules and separation of the disperse Phase as a separate layer. In an emulsion, this has broken; simple mixing fails to re-establish the stable emulsion.
2. **Creaming**-the formation of a layer of relatively concentrated emulsion.

It is a less serious type of instability than cracking because a creamed emulsion may be made homogenous again by shaking. Nevertheless, it is undesirable because the closeness of the globules in the cream favors breakdown of interface and consequent coalescence of the globules. Large droplets cream more rapidly coalesce more readily in the cream; therefore, the emulsion may eventually crack.

3. **Phase inversion**-the change of an emulsion from O/W to W/O emulsion or vice versa.

Small-Scale Preparation an Emulsion

- Good emulsions can be prepared on a small scale using a mortar and pestle although its efficiency is limited and the globule size may be larger than 10 μ m.

- Electric Mixer may also be used on a small scale although entrapped air may be a problem.
- Since the quality and stability of the product is improved by decreasing globule size and by reducing the size range of globules in the disperse phase the use of homogenizers recommended.
- When using many of the highly active synthetic agents, it is possible to obtain excellent preparations by simply adding the warmed phases to a beaker and stirring until cool. In place of a beaker, the phases can sometimes be placed into a bottle, which is then shaken.

A. Mortar and pestle: This approach is used for those emulsions that are stabilized by the presence of a multimolecular film (e.g., Acacia, tragacanth, agar, and chondrus) at the interface.

There are two basic methods.

- i) The Wet Gum (or so-called English) Method. In this method, the emulsifying agent is placed in the mortar and dispersed in water to form mucilage. The oil is added in small amounts with continuous trituration, each portion of the oil being emulsified before adding the next increment.
- ii) The Dry Gum (or so-called continental) Method. In this method, preferred by most pharmacists, the gum is added to the oil, rather than the water as with the wet gum method.

In both the Dry Gum and the Wet Gum methods, a primary, or concentrated, emulsion is first prepared using a set ratio of oil, water, and gum (emulsifying agent) from which the final product is obtained by dilution with the continuous phase.

Acacia Emulsions

Unless otherwise specified, extemporaneously prepared emulsions for internal use are made with acacia gum. To prepare acacia emulsions using a pestle and mortar, a thick (primary) emulsion must be made first. The quantities for primary emulsions have been determined by experience.

‘Parts’ is interpreted as parts by volume for fixed, mineral, and volatile oils, and part by weight for oleoresins and acacia.

Quantities for Primary Emulsions

Types of oil	Examples	Quantities for primary emulsions (Parts)			
		oil	Water	Gum (Acacia)	Tragacanth
Fixed oil	Almond oil	4	2	1	0.1
	Arachis oil				
	Castor oil				
	Cod-liver oil				
Mineral oil	Liquid paraffin	3	2	1	0.1
Volatile oil	Turpentine oil	2	2	1	0.1
	Cinnamon oil				
	Peppermint oil				
Oleo-resin	Male fern extract	1	2	1	0.1

NB. The amount of tragacanth needed is only one-tenth of the acacia amount.

When an emulsion contains two or more oily liquids the quantity of acacia for each is calculated separately and the sum of these quantities used.

The method is as follows:

1. Measure the oil very accurately in a dry measure: pour it into the bottom of a large, perfectly dry, flat bottomed mortar and allow the measure to drain by resting it against the head of a dry flat-headed pestle placed in the mortar.
2. While the oil is draining, measure the quantity of water or aqueous vehicle required for the primary emulsion in a clean measure and place this within easy reach.
3. Weigh the acacia powder.

4. Remove the oily measure from the mortar: place the gum on the oil and mix lightly for just long enough to disperse the Jumps. Take care to keep the suspension in the bottom of the mortar. Do not over mix.
5. Immediately add all the previously measured water and stir continuously but lightly in one direction until the mixture thickens under the pestle. Then triturate vigorously (i.e., Stir and mix vigorously in the mortar).
6. When the primary emulsion is well formed there is usually a characteristic 'Cracking' sound. Continue to triturate for a further 2-3 minutes to produce a white, stable emulsion. The finer the dispersion the whiter the product.
7. Gradually dilute the primary emulsion with small volumes of the vehicle ensuring complete mixing between additions.
8. Transfer to a measure and make up to final volume.
9. If necessary, decant through a muslin strainer into a calibrated bottle.

Problems:

Sometimes the primary emulsion does not form properly and the contents of the mortar become oily, thin and translucent. This is due to phase inversion, the product, which has become a W/O emulsion cannot be diluted with water and must therefore be discarded.

Possible causes are:

- Insufficient shear between the mortar base and the pestle head.
- Inaccurate measurement of water or oil.
- Cross contamination of water or oil.
- Excessive mixing of oil and gum.
- Too early or too rapid dilution of the primary emulsion.
- Poor quality acacia.

Containers and Labeling

Emulsions should be supplied in well-filled containers with airtight closures. For viscous preparations wide-mouthed amber glass bottles are most appropriate.

Labeling: shake the bottle, store in cool place but avoid freezing.

B. Other Methods

An increasing number of emulsions are being formulated with synthetic emulsifying agents. Especially of the non-ionic type.

- When using many of these highly active synthetic agents, it is possible to obtain excellent preparations by simply adding the warmed phases to a beaker and stirring until cool. In place of a beaker, the phases can sometimes be placed into a bottle, which is then shaken.
- With hand homogenizers an initial rough emulsion is formed by titration in a mortar or shaking in a bottle. The rough emulsion is then passed several times through the homogenizer.
- Kitchen-type mixers and blenders have also been used for preparation of small batches of emulsions. Care must be taken to ensure that excessive amounts of air are not whipped into the product.

Practical

1. White Liniment BP 1988

Ingredients	Master Formula	Reduced formula
Oleic acid	85ml	
Turpentine	250ml	
Dilute Ammonia Solution	45ml	
Ammonium Chloride	12.5g	
Purified Water	625ml	
Send 50ml.		

Note that 5% of the BP quantities will produce slightly more than 50ml.

Formulation: This is an ammonium soap type of emulsion. The soap is prepared by the interaction of a fatty acid (oleic acid) and ammonia during preparation of the emulsion, which is made by shaking. Ammonium oleate is an emulsifying agent but the ammonium chloride causes suppression of the ionization of the soap by common ion effect. This together with the high percentage of turpentine in the liniment causes phase inversion to occur producing a W/O emulsion.

Compounding: Mix the turpentine oil and oleic acid in a bottle 50% larger than the final volume of the product. Add an equal volume of warm water (approximately 50°C) to the dilute ammonia solution and add this solution in small quantities with shaking to the oily liquid, shaking vigorously after each addition.

Dissolve the ammonium chloride in the rest of the water and add to the bottle in small amounts. Shaking vigorously after each addition.

Transfer the appropriate quantity to a tared bottle.

Storage and shelf life: The preparation should be stored cool but not allowed to freeze.

Turpentine should be protected from light.

Container: Amber fluted bottle, preferable glass since turpentine reacts with some plastics.

Advice for patients when dispensed: shake the bottle.

The liniment should be applied to the skin with considerable friction; it should not be applied to broken or inflamed skin or near the eyes or mucous membranes.

Actions and uses: Counter irritant and rubefacient.

2. Oily Calamine Lotion, BP 1980

Ingredients	Master formula	Reduced formula
Calamine	50g	
Oleic Acid	5mg	
Wool fat	10g	
Arachis Oil	500ml	
Calcium Hydroxide solution to	1000ml	
Send 50 ml.		

Formulation: This is a 'lime' cream type of lotion. The emulgent for the arachis oil is soap produced from the calcium hydroxide and the oleic acid. Wool fat is included as an emulsion stabilizer.

Compounding: Warm the wool fat, oleic acid and arachis oil together in an evaporation basin on a water bath until the wool fat has melted. Mix well.

Sieve and weigh the calamine and place on a warm tile. Add a little of the oily mixture and rub with a large spatula until smooth.

Dilute the paste until it is fluid and transfer it as completely as possible to the basin. Stir, if necessary, with a small glass pestle to distribute the suspension evenly. Pour into a calibrated bottle previously warmed slightly on the water bath and add the lime water in small amounts, in small amounts, shaking vigorously after each addition.

Storage and shelf life: This is an official preparation but may separate on storage. It is unpreserved and therefore should not be stored for long periods. Store in a cool place but don't allow to freeze.

Container: Amber fluted bottle.

Advice for patients when dispensed shake the bottle.

Actions and uses: Soothing lotion for treatment of eczema.

3. Arachis Oil Emulsion:

Ingredients	Master Formula	Reduced formula
Arachis Oil	250ml	
Peppermint Emulsion, Concentrated	25ml	
Chloroform Water (DS)	500ml	
Water to	1000ml	
Send 40ml		

Label: Three 5ml spoonfuls to be taken three times a day with meals.

Formulation: Peppermint is included as flavoring and chloroform water as preservative.

The vehicle is prepared by using half the volume of double strength chloroform water.

Acacia is a suitable emulgent for an internal preparation.

Arachis oil is a fixed oil and the proportions for the primary emulsion are oil: water: gum-4:2:1.

Therefore the quantities for the primary emulsion are:

Castor oil.....	8 ml
Acacia in powder.....	2gm
Water q.s.....	20 ml

Compounding: The dry gum method for making acacia emulsions:

1. Accurately measure the oil and place in a large perfectly dry mortar allowing the measure of drain.
 2. Measure the water from the vehicle and have to hand.
 3. Weigh the acacia powder.
 4. Disperse the acacia powder lightly in the oil.
 5. Add the water at once and triturate until the primary emulsion is well established.
 6. The emulsion may now be diluted and the remaining ingredients added before making up to final volume.
- If insufficient water is added, or if the water is added slowly, a W/O primary emulsion will form in spite of the presence of the predominantly hydrophilic emulsifying agent (acacia).
 - Excessive amount of water also results in a poor or broken emulsion, since the viscosity of the continuous phase reduced to a point at which the oil is no longer sheared into small globules.
 - Although a W/O emulsion is favored momentarily, the well dispersed acacia particles leave the bulk of the oil phase and are adsorbed rapidly at interface where, following wetting and hydration by water, they tend to produce a stable O/W emulsion.
 - A few seconds after addition of the 2 volumes of water, a creamy white primary emulsion is formed; a cracking sound is heard as the oil is extended into filaments

and dispersed as globules in the viscous mucilage by the rapid movement of the pestle.

- The primary emulsion should be triturated for at least 5 minutes before dilution in order to produce a fine-grained product.
- Additional water, containing flavoring ingredients, preservative, and water-soluble drugs may then be incorporated.
- Oil-soluble ingredients should be added to the oil phase before emulsification is begun. The emulsion is transferred to a graduate, and portions of water are used to rinse the mortar and pestle and to bring the product to the final volume.

Storage and shelf life: A short shelf life would be appropriate for an 'unofficial' preparation.

Store in a cool place but don't allow to freeze.

Container: Amber medicine bottle with a wide mouth.

Advice for patients when dispensed: Shake the bottle.

Actions and uses: Nutritious, demulcent and mildly laxative.

4. The HLB system

- The type of emulsion, which is produced, O/W or W/O, depends on the hydrophilic-lipophilic balance, i.e., the polar-non polar nature of the emulsifier.
- Generally agents having HLB values of 3-6 produce W/O emulsion; those agents with HLB values of 8-18 produce O/W emulsions.

Fundamental to the utility of the HLB concept is the fact that the HLB values are algebraically additive. Thus, by using a low HLB surfactant with one having a high HLB it is possible to prepare blends having HLB values intermediate between those of the two individual emulsifiers. Naturally, one should not use emulsifiers that are incompatible. The following formula serves as an example.

Formulation of an O/W lotion using the HLB method

Ingredients	Formula
Liquid Paraffin	35g

Wool fat	1g
Cetyl Alcohol	1g
Emulgent	7g
Water to	100g

The required HLB values of the first three ingredients are respectively 12, 10 and 15 for an O/W emulsion.

The total percentage of the oil phase is $35+1+1=37\%$, and the proportions of the oil phase ingredients are:

Liquid Paraffin----- $35/37 \times 100 = 94.6\%$

Wool Fat ----- $1/37 \times 100 = 2.7\%$

Cetyl Alcohol ----- $1/37 \times 100 = 2.7\%$

The total 'required HLB' value is obtained as follows:

Liquid paraffin ----- $94.6/100 \times 12 = 11.4$

Wool Fat----- $2.7/100 \times 10 = 0.3$

Cetyl Alcohol ----- $2.7/100 \times 15 = 0.4$

Total HLB =12.1

Assume that a mixture of sorbitan mono-oleate (HLB value=4.3) and polysorbate 80 (HLB value=15) is to be used as the emulgent blend. The proportion of these two substances that will provide the 'required HLB' value of 12.1 is calculated as follows:

Let X=the percentage of sorbitan mono-oleate in the mixture.

Then $100-X$ = the percentage of polysorbate.

Contribution from sorbitan mono-oleate= $4.3 \times X/100$

Contribution from polysorbate 80= $15 \times (100-X)/100$

Since the total contribution must =12.1, the expression for calculating X is

$$4.3 \times X/100 + 15 \times (100-X)/100 = 12.1$$

$$4.3 + 1500 - 15X = 1210$$

$$290 = 10.7X$$

$$X=27\%$$

Hence the percentages of the emulsifying agents in the mixture are:

Sorbitan mono-oleate	27%
Polysorbate 80	73%

Since the total percentage of the mixed emulgent s in the formula is 7, the percentages of the individuals substances are:

Sorbitan mono-oleate	$7 \times 27 / 100 = 1.89$
Polysorbate 80	$7 \times 37 / 100 = 5.11$ or $7 - 1.89 = 5.11$

5. O/W emulsion

Liquid petrolatum (Required HLB 10.5)	-----50g
Emulsifying agents	-----5g
Sorbitan mono-oleate (HLB 4.3) (Span 80)	
Polyoxyethylene 20 sorbitan mono-oleate (HLB 15.0) (Tween 80)	
water qs.	-----100g
Send 50ml.	

Compounding: The oil soluble sorbitan mono-oleate (span 80) is dissolved in the oil and heated to 75°C; the water-soluble polyoxyethylene 20 sorbitan mono-oleate (tween 80) is added to the aqueous phase, which is heated to 70°C. At this point the oil phases is mixed with the aqueous phase and the whole is stirred continuously until cool.

Actions and uses: Faecal softener in treatment of chronic constipation and painful conditions of the anus and rectum.

6. Benzyl Benzoate application, BP 1988

Ingredients	Master formula	Reduced formula
Benzyl Benzoate	250g	
Emulsifying Wax	20g	
Purified Water to	1000ml	

Send 20ml.		
------------	--	--

Formulation: In this preparation 25% of benzyl benzoate is emulsified with 2% Emulsifying Wax, BP (an anionic emulgent).

Note: The benzyl benzoate is a liquid.

Compounding: First calibrate the container and place on the water bath to warm. Weigh the benzyl benzoate into an evaporating basin and put to warm. Melt the emulsifying wax and allow the mixture to cool to about 60°C.

Warm an excess of the water to 60°C. Transfer the oily mixture to the jar and quickly add (in one amount) a sufficient quantity of the warmed water up to the tare mark. It is difficult to make any additions accurately once the jar has been shaken because the preparation froths when shaken because the preparation froths when shaken.

Either replace the closure and shake until cold, or alternatively use an electric stirrer to stir gently until cold. Avoid placing the container on a cold surface until cold, or alternatively use an electric stirrer to stir gently until cold. Avoid placing the container on a container on a cold surface until fully cooled because this may cause separation of the wax.

Storage and shelf life: The preparation should be stored cool but not allowed to freeze.

Container: Amber fluted wide mouthed jar.

Advice for patients when dispensed: Shake the bottle.

The application should not be applied to broken or inflamed skin and contact with the eyes, ears, and lips should be avoided.

The patient is scrubbed with soft soap in a hot bath to open up the burrows and, immediately after drying, Benzyl Benzoate application is applied over the affected area; a second application is made on the following day. Although more than one course of treatment may be necessary, Benzyl Benzoate is more efficient, cleaner and easier to use than sulphur preparations.

Actions and Uses: Ascaricide for treatment of scabies and pediculosis.

7. Concentrated peppermint Emulsion

Ingredients	Master formula	Reduced formula
Peppermint Oil	20ml	
Polysorbate 20	1ml	
Chloroform Water (DS)	500ml	
Water to	1000ml	
Send 50ml		

Compounding: Shake the oil with polysorbate 20 and add gradually, shaking well after each addition, the DS chloroform water and sufficient water to produce 1000ml.

Actions and uses: Flavoring agent.

8. Test on Arachis oil emulsion: to determine the type of emulsion using
 - a. Dilution method.
 - b. Dye solubility method.

EXPERIMENT XVIII

Suspension

- Suspensions are disperse systems in which finely divided drug particles (called the disperse phase) are distributed throughout a vehicle (called dispersion medium or continuous phase) in which the drug exhibits a minimum degree of solubility. It includes both powders in dry form to be placed in suspension and drugs suspended in liquid vehicles.
- Suspensions can be classified into coarse suspensions in which the particles are larger than $1\mu\text{m}$ in diameter and colloidal suspensions in which the particles may be considerably less than $1\mu\text{m}$ in diameter.
- **Major applications:** Suspension dosage forms can be administered orally, injected parenterally (intramuscularly or subcutaneously) or applied externally (topically) to the skin, eye, ear, etc.
- The physical properties of suspensions and their design depend upon the nature of the dispersed phase, the dispersion medium and the pharmaceutical adjuncts/additives.
- The drug particles (dispersed phase) in suspensions could be
 - Diffusible solids
 - In diffusible solids, or
 - Poorly wettable solids

General Method of Preparation of Suspensions

1. Carefully **tare** (calibrate) the container to the required volume.

2. ***Finely powder*** any solid ingredients not already in fine powder.
3. Mix the insoluble powders in a mortar, adding first the ingredient of
4. smallest bulk and diluting it with the others in increasing order of bulk, using amounts approximately equal to bulk already in the mortar
(Geometric dilution).
5. Add enough vehicles to produce a smooth paste.
6. Add ***non-volatile solid ingredients*** dissolved in part of the vehicle and mix well.
7. Dilute with a vehicle until pourable. Examine the suspension critically and, if it contains foreign particles, strain through muslin.
(Before use, rinse the fabric with a little vehicle to detach loose fibers).
8. **Transfer it to the tared bottle** A measure is not used because complete and accurate transfer of a suspension is not possible once it has been made up to volume.
9. Add ***any volatile solid ingredients***, previously dissolved in some of the vehicle, and mix well.
10. Add any liquid ingredients; rinse the measures and mix well after each addition.
11. *Rinse the mortar and pestle* with successive volumes of the vehicle until they are quite clean, transferring the rinsing to the bottle.
12. Make up to volume with the vehicle and *shake thoroughly*.

Containers and Labeling

- Suspensions are generally packed in well-closed containers appropriate to the class of dispensed medicine be it mixture, lotion, ear drop, enema, etc.
- If the preparation is particularly viscous, a wide mouthed container (jar) may be used.
- Containers of suspensions should be of suitable size to allow shaking of the products.

- Labels for suspensions include:
 - Shake well before use
 - Store in cool place (not refrigerator)
- Also expiry dates for suspensions that should be prepared recently is one month and, if diluted, two weeks.

A. Suspensions Containing Diffusible Solids

- Diffusible solid (also called dispersible solids) are light and easily wettable solids. They readily mix with water and, on shaking, diffuse evenly through the liquid for long enough for a dose to be measured. (Ensure even distribution in each dose).
- Examples of diffusible solids are *light Kaolin, magnesium trisilicate, light magnesium carbonate, and calcium carbonate.*

1. Pediatric Kaolin Mixture, BP 1988

Ingredients	Master Formula
Light Kaolin	200 g
Amaranth Solution	10 ml
Benzoic Acid Solution	20 ml
Raspberry Syrup	200 ml
Chloroform Water (D.S.)	500 ml
Water to	1000 ml
M.Ft.M Send 50ml. Sig. 1tsp t.i.d.	

Formulation: The kaolin should preferably be sterilized to remove any contaminating soil pathogens.

Kaolin is a diffusible solid and therefore no additional suspending agent is required, although the raspberry syrup increases the viscosity of the vehicle.

Benzoic acid solution and chloroform water act as preservatives. The raspberry syrup provides a flavor and amaranth solution color, which is attractive to children.

Compounding: First tare the final container because if the preparation is made up to volume in a measure, it is difficult to transfer it satisfactorily to a bottle. The preparation should be made in a mortar of sufficient size to allow for adequate mixing of the product. Add the kaolin to the mortar and prepare a paste with the syrup and a little of the chloroform water. Add the amaranth solution and mix well.

If the dye is added at a later stage in the preparation, it will not penetrate into the powder and white specks will be visible in the final product. Dilute the suspension until pourable and transfer to the bottle. Although it is advisable to add volatile ingredients such as chloroform water to the bottle rather than the mortar, care should be taken to incorporate all the chloroform water before making up to volume with water. Add the benzoic acid solution to the bottle and make up to the final volume and shake thoroughly.

Storage and shelf life: Unless the kaolin has been sterilized, the preparation should be recently prepared.

Container: Amber medicine bottle.

Advice for patients when dispensed: Shake the bottle.

Maintain fluid intake in diarrhea. If the product is counter-prescribed the patient should seek medical advice if the condition persists.

Actions and uses: Anti diarrhea mixture for children.

2. Magnesium Trisilicate Mixture, BP 1988 (Magnesium Trisilicate Oral Suspension.)

Ingredients	Master Formula
Magnesium Trisilicate	50 g
Light Magnesium Carbonate	50 g

Sodium Bicarbonate	50 g
Conc. Peppermint Emulsion	25 ml
Chloroform Water (D.S.)	500 ml
Water to	1000 ml
M.Ft.M Send 50ml. Sig. 2tsp t.i.d.	

Formulation: Magnesium trisilicate and light magnesium carbonate act as antacids and adsorbents in the GIT. Sodium bicarbonate is a rapidly acting antacid. Peppermint provides flavoring and acts as a carminative and chloroform water acts as flavoring, preservative and vehicle. Both the magnesium powders are diffusible and no extra thickening agent is required.

Compounding: First tare the final container and select a mortar of sufficient size to allow for adequate mixing of the product. Add the sodium bicarbonate to the mortar and mix with the two insoluble powders by doubling the bulk on each addition. This method is to be preferred to making a separate solution of the sodium bicarbonate, which is time consuming. Add enough vehicles to make a smooth paste and dilute with the vehicle until sufficiently pourable to transfer to the bottle. The volatile peppermint emulsion should be added to the bottle before making up to final volume.

The mixture may also be prepared from a pre-packed mix of the three powdered ingredients with the addition of the appropriate amount of peppermint emulsion and chloroform water.

Storage and Shelf life: The mixture should be recently prepared.

Container: Amber medicine bottle.

Advice for patients when dispensed: Shake the bottle.

The mixture may be taken alone or with water or other fluid between meals. If counter-prescribed, the patient should seek medical advice if the condition persists.

Note: The mixture contains relatively high concentration of sodium ions, which may be inappropriate for some patients.

Actions and uses: Antacid used to treat dyspepsia.

3. Menthol and Eucalyptus Inhalation, BPC 1973

Ingredients	Master Formula	Scaled Quantities
Menthol	3g	1.5g
Eucalyptus Oil	8ml	4.0 ml
Light magnesium carbonate	-	2.75g*
Water to	1000 ml	50 ml
M.S.A., Send 50ml.		

Formulation: Light magnesium carbonate, a diffusible solid, is used to adsorb the volatile ingredients to ensure a uniform dispersion. The appropriate quantity is calculated:

1g for each 2g of menthol and 1g for each 2ml of eucalyptus oil

Quantity of magnesium carbonate required:

For menthol = 0.75g

For the oil = 2.00g

Total = 2.75g.

Compounding: Finely powder the menthol in a glass mortar and dissolve in the oil. Add the light magnesium carbonate in small amounts and mix well. Gradually add the vehicle to produce a pour able cream. Transfer to a previously tared container and adjust to final volume.

Storage and shelf life: The preparation is chemically stable but should be stored cool because of the volatile ingredients.

Container: Amber fluted bottle.

Advice for patients when dispensed: Shake the bottle, Not to be taken. One teaspoonful to be added to 1 pint of hot, not boiling water and the vapor inhaled for 5 to 10 minutes.

Actions and uses: Relief of nasal congestion.

4. Magnesium Sulfate Mixture, BP. 88 (Magnesium Sulfate Oral Suspension)

Ingredients	Master Formula	Reduced formula
Magnesium sulfate	400 g	
Light magnesium carbonate	50 g	
Concentrated Peppermint Emulsion	25 ml	
Double Strength Chloroform Water	300 ml	
Water FBC, sufficient to produce	1000 ml	
M.Ft.M., Mitt. 50ml		
Sig. 10 ml t.i.d., shake the bottle before use		

Formulation: Magnesium sulfate mixture is a suspension containing 40% w/v magnesium sulfate and 5% w/v of light magnesium carbonate. Both these solids act as laxatives by preventing the normal absorption of water from the intestine, with the result

that the bulky fluid contents distend the bowel, active reflex peristalsis excited, and evacuation of the contents of the intestine occur. Peppermint provides flavoring and acts as a carminative and chloroform water acts as flavoring preservative and vehicle. Light magnesium carbonate is a diffusible solid and no extra thickening agent is required.

Compounding: First tare the final container and select a mortar of sufficient size to allow for adequate mixing of the product. Mix the magnesium sulfate and light magnesium carbonate in the mortar. This method is to be preferred to making a separate solution of the magnesium sulfate, which is time consuming. Add enough vehicle to make a smooth paste and dilute with the vehicle until sufficiently pour able to transfer to the bottle. The volatile peppermint emulsion should be added to the bottle before making up to final volume.

The mixture may also be prepared from a pre-packed mix of the two powdered ingredients with the addition of the appropriate amount of peppermint emulsion and chloroform water.

Storage and shelf life: The mixture should be recently prepared.

Container: amber medicine bottle.

Advice for patients when dispensed: Shake the bottle.

Actions and Uses: A laxative.

EXERCISE

1. Write the *scaled (reduced) formula* required to prepare *magnesium trisilicate mixture* for two days.
2. Mention the advantages of suspensions over solutions and other solid dosage forms (tablets, capsules, etc.).

B. Suspensions containing indiffusible solids

- In diffusible solids will not remain evenly distributed in a vehicle long enough to ensure uniformity of dose. Examples include chalk, zinc oxide, calamine, aspirin, and phenobarbitone.
- The simplest way of correcting this problem is to increase the viscosity of the vehicle by adding a thickening agent which delays sedimentation by impeding fall of particles under gravity and by obstruction particle collision, which leads to the formation of aggregates that settle rapidly.

The thickening agents used to stabilize suspensions are substances that spontaneously form colloidal dispersions in aqueous media because of their affinity to water. There are three major groups of thickening agents:

1. Polysaccharides include acacia, tragacanth, methylcellulose, sodium alginate, microcrystalline cellulose, sodium carboxymethyl cellulose, Starch, etc.
2. Inorganic agents include bentonite, veegum, hectorite, aluminium hydroxide, etc.
3. Synthetic agents include carboxyvinyl polymer (Carbomer), colloidal silicon dioxide, etc.

Method of Preparation

Mix the thickening agent as a dry powder with the other insoluble powders in the formula. Then make a smooth past and continue as in the general method.

Alternatively, the suspending agent, previously dispersed in water to form mucilage, is levigated with the insoluble powders.

1. Phenolated Calamin Lotion, USP

Ingredients	Master formula	Reduced formula
Calamine	80g	

Zinc Oxide	80g	
Bentonite	20g	
Glycerin	20ml	
Liquefied Phenol.	10ml	
Calcium Hydroxide solution to	1000 ml	
Mitt. 50ml.		

Formulation: Calamine (basic ZnCO_3 Suitably colored with Fe_2O_3) and zinc oxide are mildly astringent and soothing to the skin. Both are in diffusible solids and therefore bentonite is included as a thickening agent which causes marked flocculation of the calamine and make it very difficult to pour from that bottle. Calcium hydroxide solution is included to cause partial deflocculation of the calamine and so make the preparation more pourable. Liquefied phenol acts as a preservative and glycerol as emollient (assists application) miscible with the aqueous vehicle.

Note: Liquefied phenol is very caustic so do not use mouth to pipette the liquid. Use a small measuring cylinder or a treated pipette.

Compounding: First, tare the final container and select a large mortar. Mix the dry powders in the mortar until evenly dispersed. It is important that the bentonite is well distributed in the in diffusible powders. Use the calcium hydroxide solution to make a smooth paste and to dilute the formed paste for transfer to the bottle. Add the glycerol and liquefied phenol to the bottle and make up to final volume with the vehicle.

Storage and shelf life: The preparation is stable, cool storage is advised because of the volatile preservative.

Advice for patients when dispensed: Shake the bottle; For external use only. The lotion is applied to the skin as required and allowed to dry.

Actions and Uses: Cooling lotion useful for treating mild sunburn.

2. Pediatric Chalk Mixture, B.P.1988 (Pediatric Chalk Oral Suspension)

Ingredients	Master formula	Reduced formula
Chalk	20g	
Powdered Tragacanth	2g	
Conc. Cinnamon Water	4ml	
Syrup	100ml	
Chloroform Water (D.S.)	500ml	
Water to	1000ml	
Mitt. 50ml.		

Formulation: Chalk is an in diffusible solid and tragacanth is included as a suspending agent. Cinnamon provides flavoring and also acts as a carminative. Syrup provides sweetening and adds to the viscosity of the product and chloroform water acts as flavor, preservative and vehicle.

Compounding: first tare the final container and select a mortar of sufficient size to allow for adequate mixing of the product.

1. Mix the chalk and tragacanth in the mortar and prepare a paste with the syrup and a little of the vehicle. Dilute until pourable and transfer to the bottle. Add the other ingredients to the bottle and shake well before making up to final volume.
2. Put the chalk in the mortar and the tragacanth mucilage in small amounts triturating to produce a smooth paste. Dilute with the vehicle until pourable and transfer to the bottle. Add the other ingredients to the bottle and shake well before making up to final volume.

Storage and shelf life: The preparation should be recently prepared.

Container: amber medicine bottle.

Advice for patients when dispensed: Shake the bottle.

Maintain fluid intake in diarrhea. If the product is counter prescribed, the patient should seek medical advice if the condition persists.

Actions and uses: Anti-diarrheal mixture for children.

Mucilages

1. Tragacanth Mucilage BPC 1973

Ingredients	Master formula	Reduced formula
Tragacanth, finely powdered	12.5	
Alcohol (90%)	25ml	
Chloroform Water	1000ml	
Mitt. 100ml.		

Formulation: The alcohol is used as a dispersing agent because it is poorly absorbed by the gum. The chloroform water acts as a preservative.

Compounding: On a small scale, the mucilage is conveniently made by shaking the ingredients together in a jar calibrated to 100ml.

Measure 95ml of the vehicle and have to hand. Put the alcohol into the jar and then add the tragacanth powder. (The order of addition is important, i.e., reverse order may lead to lumpiness).

Mix and speed the resulting suspension around the inside of the jar. Pour in the 95ml of the vehicle as quickly as possible, put on the closure and shake without delay. Success depends on speed. Any vehicle spilled on the outside of the jar can be replaced within making to volume. A product that appears lumpy at first will become homogenous on standing for a few days.

Container: Amber bottle.

Actions and uses: suspending and thickening agent.

1. Starch Mucilage, BPC 1973

Ingredients	Master formula	Reduced formula
Starch	25g	
Water to	100ml	
Mitt. 100ml.		

Formulation: Note that the mucilage is unpreserved and must be prepared as required for use in dispensed products.

Compounding: Triturate that starch with 20ml of the cold water in a small beaker or small mortar. Heat the remaining 80ml of water to boiling in a wide-mouthed conical flask or beaker. Pour the cold suspension into the center of the boiling water and reheat to boiling water with constant stirring. The resulting gelatinized mucilage should be free from lumps. Immediate rapid cooling prevents the formation of a skin. The cooled preparation may be transferred to a measure and the water lost by evaporation replaced to volume.

Storage and shelf life: the preparation should be used at once.

Advice for patients when dispensed: Not applicable.

Actions and uses: suspending and thickening agent.

C. Suspensions Containing Poorly Wettable Solids

- Some substances are both insoluble in water and poorly wetted by it. When dispersion in water are prepared, it is difficult to disperse clumps and the foam produced by shaking tends to persist because it is stabilized by the film of unwettable solids at the liquid/solid interface.
- To ensure satisfactory wetting, the interfacial energy between the solid particles and the liquid must be reduced. This may be achieved by adding a suitable wetting agent which is absorbed at the solid/liquid interface in such a way that the affinity of the particles for the surrounding medium is increased while the inter-particle forces are decreased, i.e., deflocculation occurs.
- Surface-active agents such as wetting agents for external application includes quillaia tincture and sodium lauryl sulfate. For oral administration, the less toxic polysorbates and sorbitan esters are preferred. Hydrophilic colloids such as acacia, tragacanth and the alginates will also act as wetting agents by coating

solid hydrophobic particles and imparting a hydrophilic character. These substances also cause deflocculation of particles and imparting a hydrophilic character. These substances also cause deflocculation of particles in suspension, especially at low concentration.

Method of preparation

- First, the insoluble powders are mixed with the wetting agent, using mortar and pestle. Then the vehicle is used to make a smooth paste, and the general method for the preparation of suspension is followed.

1. Sulfur Lotion, Compound, BPC 1973

Ingredients	Master formula	Reduced formula
Precipitated Sulfur	40g.	
Quillaia Tincture	5ml	
Glycerol	20ml	
Alcohol (95%) or IMS	60ml	
Calcium Hydroxide Solution to	100ml	
Mitt. 50,lm.		

Formulation: Quillaia tincture (0.5 per cent) is used as a wetting agent for the poorly wettable sulfur. Not that saponins of quillaia in aqueous solution lower surface tension,

and the extract and tincture are used as wetting agents. Glycerol is an emollient and with the alcohol, Industrial Methylated Spirit (IMS) may be used in place of the alcohol.

- In unofficial sulphur lotion, 0.25 per cent sodium lauryl sulfate may be used instead of quillaia tincture.
- The dispersion formed sediments quickly but can be shaken into a suspension from which accurate aliquots may be removed by the patient.

Compounding: Mix the quillaia tincture, alcohol and glycerol and triturate the mixture with the sulfur in a mortar. Gradually dilute with the limewater and transfer to the previously tarred container to make up to final volume.

Storage and shelf life: The preparation is stable.

Container: Amber fluted bottle.

Advice for patients when dispensed: Clean the skin before use and apply sparingly to the affected area. The treatment should be discontinued if excessive dryness or irritation occurs.

Actions and uses: Mild antiseptic used in treatment of acne.

PHARMACEUTICS II LABORATORY MANUAL

TABLE OF CONTENT

EXPERIMENT 1 Ointment.....	
EXPERIMENT 2 Ointments (cont.)	
EXPERIMENT 3 Ointments (cont.)	14
EXPERIMENT 4 Pastes.....	16
EXPERIMENT 5 Jellies.....	18
EXPERIMENT 6 Creams.....	20
EXPERIMENT 7 Suppositories.....	23
EXPERIMENT 8 Suppositories (Cont.).....	28
EXPERIMENT 9 Powder dosage forms.....	34
EXPERIMENT 10 Granule dosage forms.....	37
EXPERIMENT 11 Effervescent granules.....	39
EXPERIMENT 12 Incompatibility.....	45

EXPERIMENT 1

OINTMENTS

- Ointment are semisolid preparations intended for external application to the skin or mucous membranes
- Ointments may be medicated or non-medicated, the later type being commonly referred to as ointment bases and are used as such for their emollient or lubricant effect or used as vehicles in the preparation of medicated ointments
- There are four main classes of ointment bases
 1. Hydrocarbon Bases
 2. Absorption Bases
 3. Water-Miscible Bases
 4. Water Soluble Bases

Preparation

- Since ointments are applied primarily to irritated areas, it is an unbreakable rule that these preparations must contain no granular or gritty particles, which might cause further irritation
- Regardless of the condition of the skin, ointments must be smooth and free from granular or gritty particles
- Three types of equipment are available for the preparation of ointments:
 - a) Ointment slab and Spatula
 - b) Mortar and Pestle
 - c) Ointment mill
- Two techniques are used in mixing ointments

1. Mixing by fusion

- On a small scale, fusion is usually carried out in an evaporating basin over a water bath
- Fusion is necessary when waxes, hard paraffin, fatty alcohols, fatty acids or any hard wax-like material is included in the formula
- Fusion is used also when a medicament is soluble in the melted base

2. **Mixing by triturating**

- This method is applicable when the medicament is a solid insoluble in the base or a liquid present in small amounts
- Triturating may be carried out with an ointment tile and spatula or using a mortar and pestle

Packaging

- Ointments are dispensed in either ointment jars or collapsible tubes. However, ointments prepared extemporaneously are usually packed in ointment jars.
 - Procedure will be demonstrated
 - The ointment should be packed tidily and the two surfaces should be smoothed nearly
- Ointment jars should be closed tightly when not in use, particularly if the ointment contains water to prevent loss through evaporation

Storage

- Ointments should be stored in a cool place to prevent softening and eventual liquefaction of the base
 - Insoluble medicaments may settle to the bottom of the container if the ointment is permitted to liquefy
 - Excessive heat or cold may cause an emulsion base to separate into the two phases

Practical

1. **Determination of Water:**

- Water number is the largest amount of water in grams that can be incorporated in 100grams of an ointment base at normal temperature (about 20°)
- Determine the water number of
 - (1) Pure soft petroleum: take 10gram
 - (2) Soft petrolatum (95%) + anhydrous lanolin (5%): add 0.6g anhydrous lanolin to the 10g soft paraffin in (a)
 - (3) Soft petrolatum (85%) + anhydrous lanolin (15%): add 1.2g anhydrous lanolin to the mixture in (b)

Procedure

Gradually add water in drops to the base contained in a mortar from a burette with constant titration and stop adding the water when it could no longer be incorporated into the base. Record the amount of water incorporated into each of the three bases, (a) to (c) and comment on the result

2. Sulphur Ointment using Simple Ointment BP

a. Simple Ointment BP

Ingredients	Master formula	Reduced Formula
Wool fat	50 g	-
Hard paraffin	50 g	-
Cetostearyl alcohol	50 g	-
Yellow or White soft paraffin	850 g	-
Send 25g		

Formulation: Simple Ointment is an absorption base prepared by fusion. Yellow or white soft paraffin is chosen according to the color of the finished product. In this case, since there is no dark colored ingredient to be added, white soft paraffin should be used

Compounding: Grate the hard paraffin and weigh out the ingredients for the base. Place to melt in an evaporating dish on a water-bath. Stir gently to aid melting and to mix the ingredients. When homogenous, remove from the heat and continue to stir gently until cold

- **NB.** Weigh the wool fat and the soft paraffin using waxed papers to allow full recovery of the materials

Storage: store in a cool place

Action and uses: Ointment base emollient

b.Sulfur Ointment, BP

Ingredients	Master formula	Reduced Formula
Precipitated sulfur, finely sifted	100 g	
Simple ointment	900 g	
Send 25 g		

Formulation: The BP directs that the simple ointment be prepared with white soft paraffin. 25g of simple ointment is adequate to ensure that 22.5g will be available (to compensate for the manipulative losses)

Compounding: Weight out the amount of precipitated sulfur required and place it on the center of the ointment tile; weight the required amount of simple ointment (using a piece of waxed paper), and place it on one side of the tile. Take a portion of the base (about two to three times the weight of the power) and levigate it with the sulfur until there is no sign of any gritty particles (This can be checked for by spreading a thin layer on the tile). Gradually incorporate the remaining base in portions geometrically and continue levigation until uniformly mixed. Finally, collect the ointment together to the center of the tile using the spatula and pack into an ointment jar

Storage and Shelf life: Store in a cool place. An expiry date of 4 weeks is appropriate.
For external use only

Container: Wide-mouthed amber jar with a greaseproof paper disc

Advice for patient when dispensed: The skin should be cleansed prior to application and the ointment applied sparingly to the affected area

Action and uses: Mild antiseptic used in the treatment of acne and scabies

Assignment

1. Calculate the water number for the three ointment bases and comment on the result, i.e.
 - Explain why the amount of water incorporated into the base increase as the amount of Anhydrous Lanolin increase?
 - What is the composition of Anhydrous Lanolin and which fraction is responsible for increasing the water number?
 - What is the practical application of this?

Experiment 2

OINTMENTS (cont)

1. Whitfield's Ointment using Emulsifying Ointment

a. Emulsifying Ointment, BP

Ingredients	Master formula	Reduced formula
Emulsifying Wax	300 g	
White Soft Paraffin	500 g	
Liquid Paraffin	200 g	
Send 25 g		

Compounding: Place ingredients of the base in an evaporating dish on a water-bath and allow melting. It is most convenient to weigh the liquid paraffin directly into the dish and to add the other ingredients. Stir gently to aid melting and to mix the ingredients. When homogenous, remove from the heat and continue to stir gently until cool

b. Compound Benzoic Acid Ointment, BP (Whitfield's Ointment)

Ingredients	Master formula	Reduced formula
Benzoic Acid, in Fine powder	60 g	
Salicylic Acid, in fine powder	30 g	
Emulsifying Ointment	910 g	
Send 25 g		

Compounding: Sift the medicaments through separate 180µm sieves. Weigh the powders after sifting and mix them. Then, place them on an ointment tile and levigate with two or three times their weight of the base until homogenous. Gradually incorporate the remaining base.

- **Note:** Salicylic acid irritates the respiratory tract, so precautions should be taken to avoid inhalation

Storage and Shelf life: Store in a cool place

Container: Wide-mouthed amber jars with a greaseproof paper disc

Advice for patients when dispensed: The ointment should be applied sparingly to the affected area. It should not be applied to the broken or inflamed skin

Action and uses: Topical antifungal preparation

2. Methyl salicylate ointment, BP 1980

Ingredients	Master formula	Reduced formula
Methyl salicylate	500g	
Beeswax	250g	
Hydrous wool fat	250g	
Send 25 g		

Formulation: Methyl salicylate is a volatile liquid and should be added to the base at as low a temperature as possible. Beeswax helps to stiffen the base, which would otherwise be very soft by the large proportion (50%) of liquid methyl salicylate

Compounding: Melt the grated beeswax and hydrous wool fat in an evaporating dish on a water-bath. Remove from the heat and cool until about to solidify. Add the methyl salicylate and stir until cool

Storage and Shelf life: Store in cool place

Container: Wide-mouthed amber glass jar with a greaseproof paper disc and well-fitting closure

- **Note** that methyl salicylate reacts with some plastics

Advice for patients when dispensed: The ointment should be massaged well into the skin. It should not be applied to broken or inflamed skin or near to the eyes or mucous membranes. The container should be kept tightly closed

Action and Uses: Relief of pain in rheumatic conditions

Experiment3

OINTMENTS (Cont)

1. Calamine and coal tar ointment

Ingredients	Master formula	Reduced formula
Calamine, finely sifted	125g	-
Zinc Oxide, finely sifted	125g	-
Strong Coal Tar Solution	25g	-
Hydrous Wool Fat	250g	-
White Soft Paraffin	475g	-
Send 25g		

Formulation: The ointment contains two insoluble solids and a volatile liquid

Compounding: Sift the zinc oxide and calamine through separate 180µm sieves and weigh the required amounts. Melt the soft paraffin and hydrous wool fat at as low a temperature as possible. Add small amounts of each powder alternately to the molten base and stir thoroughly until homogenous. Continue to stir until the temperature falls to 40°C before adding the strong coal tar solution and stirring until cold

- Note that because coal tar solution is volatile, it should not be weighed in advance and left uncovered on the bench to evaporate

Storage and shelf life: Store in a cool place

Container: Wide-mouthed amber jar with a greaseproof paper disc

Advice for patients when dispensed: The ointment should be applied sparingly to the affected area. It should not be applied to the broken or inflamed skin or near the eyes. It stains skin, hair and fabrics

Actions and uses: Mild astringent and anti-pruritic

2. Ichthammol Ointment

Ingredients	Master formula	Reduced formula
Ichthammol	100g	-
Wool fat	450g	-

Yellow Soft Paraffin
Send 20 g

450g

-

Formulation: Ichthammol is an almost black viscous liquid with a strong tarry odor. It consists of the ammonium salts of the sulfonic acids. It is water soluble. Incorporation in an ointment is facilitated if the base contains a large amount of wool fat, 45% in this case

Compounding: Place the ingredients for the base in an evaporating dish on a water bath and allow it to melt. Stir gently to aid melting and to mix the ingredients. When homogenous, remove from the heat and cool to about 40°C. Add the ichthammol and stir until cool

Storage and shelf life: Store in a cool place

Container: Wide-mouthed amber jars with a greaseproof paper disc

Advice for patients when dispensed: The ointment should be rubbed lightly on the affected area. It should not be applied to the broken skin

Actions and uses: Mild antibacterial and anti-inflammatory

Experiment 4

PASTES

- Pastes are ointment like preparations intended for external application
- Most pastes differ from ointments in containing large amounts of finely powdered solids
 - Thus, they are usually stiffer than ointments, less greasy and more absorptive
- They are applied generously, either with a spatula or spread on lint or other dressing and are not removed for long periods
- When applied to the skin they adhere well, forming a thick coat that protects and soothes inflamed and raw surfaces and in particular, minimizes the damage done by scratching in itchy conditions such as chronic eczema
- Most pastes are unsuitable for treating hairy parts such as the scalp because they are very difficult to remove from the hair as they form a densely matted mass

Compounding and Packaging

- For all practical purposes, pastes are prepared in the same manner as ointments
- Pastes are generally packaged in glass ointment jars. They are often too viscous for easy extrusion from tubes

Practical

1. Resorcinol and Sulfur Paste, B.P.C

Ingredients	Master formula	Reduced formula
Resorcinol, finely sifted	50g	
Precipitated Sulfur	50g	
Zinc Oxide, finely sifted	400g	
Emulsifying Ointment	500g	
Send 20 g		

Compounding: Sift the powders through separate 180µm sieves. Weigh the powders after sifting and place on a slightly warm tile and gently mix. Levigate the powders with little emulsifying ointment until smooth. Then dilute gradually

Storage: Store in a cool place

Container: Wide mouthed amber jar with greaseproof paper disc

Advice for patient when dispensed: The paste should be applied liberally on the affected area

Actions and Uses: Treatment of acne

2.Zinc Oxide Paste, U.S.P. 1990 (Syn. Lassar's Plain Zinc Paste)

Ingredients	Master formula	Reduced formula
Zinc Oxide, finely sifted	250g	
Starch, finely sifted	250g	
White Petrolatum	500g	
Send 20g		

Compounding: Melt the base with a minimum of heat and sift the powders through separate 180µm sieves. Mix the required weights of powders in a warm mortar. Triturate with a little melted base until smooth, gradually adding the rest of the base mix until cooled

- **NB:** The paste is much easier to manipulate in a warm mortar

Storage: Store in a cool place

Container: Wide-mouthed amber jar with greaseproof paper disc

Advice for patient when dispensed: The paste should be spread thickly on white lint and applied to the affected area

Action and Uses: Absorbent, protectant and mild astringent used for variety of dermatological conditions including eczema and psoriasis either alone or with additional medicaments

Experiment 5

JELLIES

- Jellies are transparent or translucent, non-greasy, semisolid gels generally applied externally
- Jellies are used for medication (Medicated jellies), lubrication (Lubricating jellies) and some miscellaneous applications

Formulation of Gels

- Gelling agents are usually organic hydrocolloids but, occasionally, hydrophilic inorganic substances
 - **Organic hydrocolloids:** Tragacanth, Sodium alginate, Pectin, Starch, Gelatin, Cellulose derivatives (Methylcellulose, Sodium carboxymethylcellulose, Hydroxypropyl methylcellulose), Carbomer, Polyvinyl alcohols, and others
 - **Hydrophilic inorganic substances:** Clays like bentonite
- Other additives for gels
 - **Humectants:** Loss of water can quickly lead to coarsening of the matrix and humectants such as glycerol, propylene glycol or Sorbitol Solution may be added to retain water
 - **Preservatives:** Gels have high water content and therefore are liable to support microbial growth unless a suitable preservative is added
 - **NB** The chosen preservative must be compatible with the gelling agent
 - **Chelating agents:** Bases and medicaments sensitive to heavy metals are sometimes protected by chelating agents such as EDTA
- **Packaging**
 - Lubricating jellies should be packed in collapsible tubes. However, stiffer gels for application to the skin may be supplied in ointment jars
 - Containers should be well filled and airtight to prevent evaporation
- **Storage**
 - Jellies should be stored in a cool place to prevent loss of water via evaporation

Practical

1. Lubricating Jelly with Sodium Carboxy Methyl Cellulose (Na C.M.C.)

Ingredients	Master formula	Reduced formula
Na C.M.C. (Medium Viscosity Grade)	5.0g	
Glycerol	15.0g	
Methyl Hydroxybenzoate	0.1g	
Patent Blue V	0.001g	
Purified Water, FBC to Send 40g	100.0g	

Formulation: Sodium carboxymethylcellulose is the gelling agent. Methyl hydroxybenzoate acts as a preservative and glycerol as humectants. Although advisable, it is not essential for rectal lubricants to be sterile but, if they are not, it is useful to include a dye, like patent blue V, as a warning that the preparation is not sterile and so unsuitable for lubricating articles used in sterile regions of the body. A label to this effect is also desirable

Compounding: Dissolve the methyl hydroxybenzoate in the glycerol and add the sodium carboxymethyl cellulose. Heat the water to boiling, add the glycerol mixture to it and stir until a clear gel is formed. Then add the dye solution, adjust to weight and stir until cold

Storage: Store in a cool place

Container: Collapsible metal tube

Advice for patients when dispensed: Not applicable

Actions and uses: Non-sterile lubricating jelly for lubricating rubber gloves for rectal examination

Experiment 6

CREAMS

- Creams are semisolid emulsions of either the oil-in-water or the water-in-oil type, usually medicated, intended for external application
- Creams are usually employed as emollients or as medicated applications to the skin
- Creams are pharmaceutically and cosmetically appealing as they are soft, easy to apply, cooling to the skin, and many are water-removable

Preparation

- Creams may contain one or more medicaments in solution in one or other of the phases. Finely powdered insoluble medicaments may also be dispersed in a cream base
- Generally, creams are prepared as follows
 - The components of the oily phase (usually including the emulgent) are heated until molten and the temperature rises to 75°C
 - The components of the aqueous phase are mixed in a separate vessel and also heated to 75°C
 - The aqueous phase is then added to the oily phase at the same temperature and the resulting emulsion is stirred gently until cool
 - Thermometer should be used to control the temperatures of the two phases
 - Rapid cooling may result in separation of high melting point components
 - Excessive aeration caused by vigorous stirring may also lead to a granular product

Packaging

- Wide-mouthed jars may be used for creams where the risk of contamination in use is considered to be minimal, e.g. oily creams. The containers must be well closed and prevent water evaporation. The mouth of the jar should be covered with a disc of greaseproof paper
- Collapsible metal or flexible plastic tubes, however, are to be preferred since these reduce the risk of contamination in use
 - **NB.** Most proprietary cream products are packed in tubes

Storage

- Creams should be stored in a cool place but should not be allowed to freeze

Practical

1. Bees Wax-Borax Cold cream

Ingredients	Master formula	Reduced formula
Purified Water	24.0g	
Borax	1.19g	
Olive Oil	50.0g	
Bees Wax	12.5g	
White Soft Paraffin	12.5g	
Make cream, send 20g		

Formulation:

- Cold cream is a water-in-oil emulsion

- The cream is self-emulsifying i.e., the emulgent is a product of reaction between borax and free fatty acids of the oil and the wax. Accordingly, the amount of alkali (borax) should be calculated
- Calculate the amount of borax required for the above formula

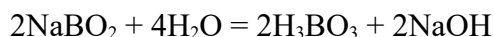
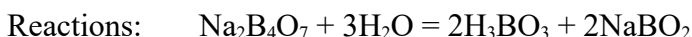
$$\text{Mol. Wt. Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} = 381.43$$

$$\text{Mol. Wt. KOH} = 56.11$$

$$\text{Mol. Wt. NaOH} = 40$$

$$\text{Acid value of Olive Oil} = 2$$

$$\text{Acid value of Bees Wax} = 20$$



- Therefore, 1 mole $\text{Na}_2\text{B}_4\text{O}_7$ produces 2 moles NaOH
 - **Acid value** is the number of milligrams of potassium hydroxide needed to neutralize the free fatty acids in 1 gram of fat or oil

Compounding: Melt the beeswax and soft paraffin on a water bath. Add the olive oil and continue heating to 75°C. Dissolve the sodium borate (borax) in the purified water and warm to the same temperature. Then, gradually add the aqueous phase to the oil phase with gently stirring. Maintain the temperature for about 10 minutes while stirring and continue the stirring until cold

Storage: Store in a cool place but don't allow to freeze

Container: Collapsible metal tube or wide mouthed jar

Action and Uses: Emollient cream base

- **Note:** the cooling effect in cold cream is caused due to the slow evaporation of water present in the internal phase

2. Vanishing Cream

Ingredients	Master formula	Reduced formula
Purified Water	79.0g	
Stearic Acid	15.0g	
Potassium Hydroxide	1.04g	
Glycerin	5.0g	

Make cream, send 30g

Formulation

- Vanishing cream is an oil-in-water emulsion. A part of the stearic acid is saponified (15-35%) with the alkali (KOH) and the resulting soap emulsifies the bulk of the acid
- Glycerin is added as a humectants to reduce the rate of drying out
- Calculate the Amount of KOH needed to Neutralize 35% of the stearic acid?

Mol. Wt. Stearic Acid = 284.5

Mol. Wt. KOH = 56.11

- **Note:** Stearic acid $[\text{CH}_3(\text{CH}_2)_{16}\text{COOH}]$ is a monocarboxylic acid. Therefore, 1 mole KOH reacts with 1 mole Stearic Acid

Compounding: Melt and heat the stearic acid to 75°C. Dissolve the potassium hydroxide in the purified water, add the glycerin and warm to the same temperature. Then, gradually add the aqueous phase to the oil phase with slow stirring maintaining the temperature for at least 10 minutes and continue the stirring until cold

Storage: Store in a cool place but don't allow to freeze

Container: Collapsible metal tube

Actions and Uses: Hand or foundation cream

- **Note:** 'Vanishing Creams', because it leaves a very matt (dull), non-greasy residue on the hand to the extent that the cream appears to vanish during application

Experiment 7

SUPPOSITORIES

- Suppositories are solid medicated preparations designed for insertion into the rectum to exert local or systemic effect
- Pessaries are similar solid medicated preparations designed for insertion into the vagina, usually to exert local effect

Formulation

- There are two main classes of suppository base
1. **Fatty bases:** designed to melt at body temperature
 - Examples: Theobroma oil (cocoa butter), synthetic fats (Witepsols)
 2. **Water-soluble or water-miscible bases:** designed to dissolve or disperse within the body
 - Examples: Glycerogelatin bases, macrogol bases

Compounding

Suppository and Pessary moulds

- The normal capacities of commonly used moulds. 1g (15 gr), 2g (30 gr), 4g (60 gr), and 8g (120 gr)

- The nominal capacity of mould implies a calibration using oil of theobroma and therefore will vary with different bases depending upon their density
 - **NB.** The capacity of moulds should be confirmed by filling each mould with base alone, weighing the products and taking the mean weight as the true capacity

Displacement Value

- **Displacement value:** the number of parts by weight of medicament that displaces one part by weight of the base
- Displacement values for fatty bases are similar to those for theobroma oil
 - **NB.** When the medicament is prescribed as a percentage, the displacement value is irrelevant, to use it would make the percentage incorrect

Mould Lubrication

<u>Base</u>	<u>Lubricant</u>
Theobroma	Soap spirit (soft soap 10g), glycerol (10ml) and 90% alcohol (50ml)
Glycero-gelatin bases	Liquid paraffin or Arachis oil
Synthetic fats	No lubricant required
Macrogol bases	No lubricant required

- **NB.** The chosen lubricant must be immiscible with the base and compatible with the medicament and any other formulation additives

Preparation of Suppositories

- **Two methods mainly are used:**
 1. **Compression or cold process:** the drug is mixed with shredded vehicle and the mixture then is forced into a mould and subsequently extruded
 2. **Fusion or hot process:-** the drug is incorporated into the melted base and the mass is poured into the mould just before it congeals (Principal method for the preparation of suppositories)

Preparation of Suppositories with Fatty Base

1. Calculate the required amounts
 - It is usual to prepare for at least two more suppositories than are ordered to compensate for the manipulation losses
2. Prepare the mold: Select a clean dry mould and place it on a clean tile

- The mould should be carefully washed in warm, soapy water and thoroughly dried
 - Care should be taken not to scratch the internal surface
 - Make sure that the two halves fit correctly to ensure that there is no leakage of material
3. Prepare the base. Weigh shredded base into an evaporating basin
 - Shredding the base speeds melting and helps to avoid over-heating
 - Porcelain basins are to be preferred for fatty bases since their relatively poor heat conductivity reduces the risk of over-heating
 4. Prepare the medicament: Insoluble solids should be powdered if necessary and sieved before weighing (180µm). Other ingredients should be weighed or measured
 5. Melt the base. Place the dish containing the shredded base on a water bath and heat until about two-thirds of the content has melted. Remove from the heat and stir gently with a small spatula to allow the residual heat in the dish to melt the remainder of the base. (Overheating may occur if the base is left over the heat until completely melted)
 6. Incorporate the medicament

Insoluble solids

- Heat a small tile until it is comfortably warm to the hand
- Mix the powders on the tile with a flexible spatula
- Pour about half of the melted base on to the mixed medicaments and work into a smooth dispersion as quickly as possible by levigating with the spatula. If the base solidifies it can be softened by holding the tile over the water bath for a few seconds
- Transfer the dispersion to the dish, leaving virtually none on the tile, and stir to form a homogenous mixture

Soluble solids, semisolids and liquids

- Incorporate directly into the molten base and stir to form a homogenous mixture
7. Fill the mould: The homogenous mixture in the dish should be stirred until the base begins to thicken. Then fill each cavity in the mould to *overflowing* to allow for contraction on cooling. If the consistency of the base is correct, the 'overflow' should remain above each cavity. If setting occurs during pouring, the base can be very briefly re-heated to restore flow
 - **NB** The base should be stirred throughout to prevent sedimentation of insoluble solids
 8. Remove the excess. When the mass has set (2-3 minutes under normal circumstances), contraction of the base will have occurred and the excess may be removed. Press the blade of a firm spatula on the excess at one end of the mould

and draw it across the mould pressing down at the same time. This process will tend to fill any holes in the suppositories, which may have resulted from pouring too cold

- Reserve the excess together with the surplus mass in case there is a need to re-melt and re-pour any failures
9. Open the mould: After 10 minutes standing in a cool place the suppositories may be removed from the mould, inspected and packed

Additional Precautions for Theobroma Oil Suppositories

1. Thoroughly lubricate the mould: Lubricant should be applied to all the inner surfaces including the flat faces of the mould (Do not use cotton wool because it leaves fibers)
 - After lubrication, close the mould and invert it on a clean paper or tile to allow any excess lubricant to drain and cool it in a refrigerator or icebox
 - **NB:** Cooling hastens setting of the base and reduces sedimentation of suspended solids
2. **Take particular care not to overheat the base** (polymorphism)
 - If melted at not more than 36°C and slowly cooled, it forms stable beta (β) crystals with normal melting point
 - If over-heated, it may produce, on cooling, unstable gamma (γ) crystals which melt at about 15°C or alpha (α) crystals melting at about 20°C
3. After pouring put the mould in an icebox or refrigerator and leave it for about 30min
4. If, through inadequate lubrication or though pouring too hot, the products adhere to the mould on opening, push them out by their flat ends, to avoid damaging the tips

Shelf life: Most suppositories and pessaries are stable preparations provided that the packaging provides adequate protection and that the storage temperature is low

Containers: Extemporaneously prepared suppositories and pessaries should be individually wrapped in metal foil or waxed paper (if the medicament interacts with metals). Glass or plastic screw-cap jars are suitable containers, particularly for products that are hygroscopic

Special Labels and Advice

- Store in a cool place
- For rectal use only (suppositories) or for vaginal use only (pessaries)
- If appropriate, the patient should be advised to unwrap the product before use

Practical

1. Displacement Value

- Method for determining displacement value: Using nominal 1g mould
 - Prepare and weigh six suppositories of theobroma oil (unmedicated base)
= A g

- Prepare and weigh six suppositories containing 20% Zinc Oxide (medicament) = B g
 - Calculate the amount of base ($C_g = 80\%$ of B_g) and medicament ($D_g = 20\%$ of B_g) in the six medicated suppositories
 - Amount of base displaced by the medicament = $A - C$
 - Therefore, Displacement Value (D.V.) = $D_g / A - C$
 - **NB:** Literature value for the D.V. of ZnO is 5.0
2. **Quantity of base required** in a prescription
- Base displaced by the drug in the prescription = Drug in the prescription/D.V
 - Quantity of base required in the prescription = Weight of unmedicated suppositories – Base displaced by the drug in the prescription
3. **Chloral Hydrate Suppositories**

Ingredients Formula

Chloral hydrate 60mg

Theobroma oil Sufficient to fill a 1g mould

Send 6 suppositories

Calculation: Prepare for 10 suppositories to allow for manipulation losses

D.V. Chloral Hydrate = 1.5

Formulation: The amount of medicament is not enough to cause serious lowering of m.pt but the product is most easily made with a base of elevated m.pt

Compounding: Use the method for preparation of suppositories with fatty base

- Finely powder the medicament before weighing to hasten dissolution
- Carefully melt the base and cool it to the consistency of thick cream. Add the finely powdered chloral hydrate and stir until fully dissolved. Complete the succeeding stages in the usual way
- Make sure the suppositories have fully set before opening the mould

Storage: store in cool place

Container: Amber glass screw capped jar or plastic pot

Advice for patients when dispensed: After unwrapping, insert the suppositories as high as possible to the rectum

Label: One to be inserted at night

Actions and Uses: Hypnotic

Assignment

1. Calculate the DV of zinc oxide from Practical “1”?
2. Calculate the quantities required to make ten medicated suppositories with theobroma oil each of which containing 200mg of drug A (DV=3.0) 60mg drug B (DV=1.5), 120mg of drug C(DV=5.0) and 60mg of drug D(DV=1.0). Assume the use of 2gram moulds?

Experiment 8

SUPPOSITORIES (Cont)

(Water-Soluble and Water-Miscible bases)

Preparation of suppositories with macrogol bases

- These are prepared in essentially the same way as the fatty base suppositories with special attention to the following
 - a. The mould must not be lubricated or chilled
 - b. The constituent macrogols are melted together, and if the base contains water, this should be warmed separately to the same temperature, added to the melted base and the mixture stirred
 - c. Water-soluble ingredients are dissolved in the water before added. Insoluble ingredients are incorporated by the method described for fatty bases
 - d. The mass is poured as cool as possible and the mould overfilled to allow for significant contraction on cooling

Preparation of suppositories with a glycerol-gelatin base

1. Calculate the quantities required
 - Glycerol Suppositories mass is 1.2 times as dense as theobroma oil. Multiply the total quantity of base required to fill the nominal capacity by 1.2 to take into account the greater density
 - When appropriate, displacement values are used in the normal way and after the amount of base has been calculated it is multiplied by the density factor 1.2
 - Prepare excess to allow for manipulation losses
2. Prepare the mould. Lubricate the mould, including the internal flat faces, using liquid paraffin or arachis oil. Then invert and cool it in a refrigerator or icebox
 - **NB.** Lubrication of the faces prevents sticking if melted base seeps between them

3. Prepare the medicament. Thermostable soluble medicaments should be dissolved in the water before heating. Insoluble solids should be sifted (180 μ m)
4. Prepare the base
 - Weigh the glycerol into a suitable dish and heat to 100°C, but not appreciably higher, and keep it hot on a water bath
 - Weigh the water into a second tared dish (the weight of the dish will be required for adjustment of the product to weight). Use approximately 30ml per 100g of base
 - **Note:** Although the calculated amount of water for 100g of Glycero-gelatin base is 16g, the use of an excess helps the dissolution of the gelatin and it can easily be adjusted if necessary by evaporation
 - Bring the water to boiling and remove it from the source of heat to prevent undue evaporation
 - Add the gelatin to the water taking care to avoid the sides of the dish and stir gently to wet the material thoroughly. Preferably use powder gelatin
 - Add the hot glycerol to the dish, re-stir and place it on a boiling water bath for 15 minutes or until solution is complete. Make sure that the dish is low in the bath for maximum heating. Stir often but gently to avoid incorporation of air bubbles
5. Heat treatment of the base: Although pharmaceutical grades of gelatin should not contain any pathogens, the base may, as a precaution, be heat-treated by maintaining it at a temperature of 100°C for 1 hour in an electric steamer
 - This is particularly important for products that are intended for application to abraded skin or to the vaginal mucosa
 - The heat treatment is carried out before adjustment to weight and before the addition of thermolabile ingredients
6. Adjustment of base to weight. When solution is complete, weigh the dish and contents and, if necessary, evaporate or add hot purified water to obtain the correct weight
 - After adjustment to weight, any large air bubbles can be removed by passing the molten mass through a warm muslin strainer into a second hot dish
7. Incorporate any medicament: Soluble thermolabile medicaments are dissolved in a little water before adding to the molten mass. Insoluble substances are rubbed down on a tile with part of the glycerol
 - **NB.** The melted base is not used for this purpose because it is much more difficult than a fatty base to re-melt if it solidifies
8. **Fill the mould:** Watch carefully for any tendency of the suppositories to split in the centerline and, if this is seen, invert the mould and try to open it from the bottom. After removal from the mould and inspection, the products may be very

lightly lubricated with liquid paraffin before packing to prevent sticking in the container

Practical

1. Glycerol suppositories BP 1988

Ingredients	Master Formula	Reduced formula
Gelatin	14g	-----
Glycerol	70g	-----
Purifiedwater	q.s (to produce100g)	-----

Send 6 child's size suppositories

Calculation

- The usual size of a glycerol suppository for a child is 2g (1g for infants and 4g for adults)
- Prepare sufficient base for 10 suppositories to allow for relatively large manipulative losses
- The quantity of mass for 10 suppositories is
 $10 \times 2 \times 1.2 = 24\text{g}$
- In this case make 25g because that is a more convenient fraction of the master formula quantities
- 100 gram of base require 30ml of water, 25gram of base, therefore, require
 $30 \times 25 / 100 = 7.5\text{ml}$ of water initially

Compounding: Add the gelatin to the purified water heated nearly to boiling, then add the glycerol previously heated to 100°C and heat the mixture on a water bath for 15 minutes or until solution is complete. Adjust the weight of the product by the addition of hot purified water or by evaporation on a water bath. Complete the succeeding stages in the usual way

Storage: Store in a cool place

Container: Amber glass screw-capped jar or plastic pot

Advice for Patients when dispensed: After unwrapping, moisten with a little water and insert the suppository as high as possible into the rectum

Actions & Uses: Promotes peristalsis and bowel evacuation in the treatment of constipation

2. Macrogol base (II)

Ingredients	Master formula
Macrogol 1000	96g
Macrogol 4000	4g

Send three 2g suppositories

Compounding: Use the general method for preparation of Macrogol Suppositories

Assignment

1. Calculate the quantities required to make 15 glycono-gelatin suppositories each of which containing 200mg of drug A (DV=1.1) and 600mg of drug B(DV=5.0). Assume the use of 2g moulds?

Experiment 9

POWDER DOSAGE FORMS

- Powders may be defined as dry mixtures of pulverized drugs and/or chemicals, which are used internally or externally
- Depending upon their intended use, powders are packaged and dispensed in two ways, as bulk powders or as divided powders
 - Bulk Powders
 - The substances prescribed in this form are bulky powders such as light and heavy magnesium carbonate, magnesium trisilicate, etc.
 - It is when accuracy of dosage is not critical, that powders may be dispensed in bulk
 - Divided Powders
 - This form of powder may contain one or more active ingredients together with an inert diluent to produce a minimum quantity (120mg) that can be weighed by the dispenser and handled by the patient
 - After the powder blend is prepared, it can be subdivided into the required number of doses by weighing or by using the block and divide method
 - The papers in which divided powders are traditionally folded are called powder papers
 - The paper of choice for hygroscopic or deliquescent powders is waxed paper because of its waterproof qualities. Also powders containing volatile ingredients should be similarly wrapped

Method for wrapping divided powders: Demonstration

Compounding

- The basic techniques for the compounding of powders are weighing Size Reduction and Mixing
- On a small scale the powdered ingredients may be mixed with a mortar and pestle using the *geometric dilution* method

Packaging:

- For bulk powders
 - For internal powders: Wide-mouth screw cap glass jars

- A 5ml spoon should be supplied for measuring dose
- For dusting powders: Colored glass or plastic jars with a recloseable perforated lid
- For Divided powders
 - Folded powder paper packets, Aluminium foils, Heat-sealed plastic bags

Storage and labeling:

- Store in a cool, dry place
- Powders intended to be used externally or locally should always bear “for external use” or “For local use” auxiliary labels

Practical

1. Compound Magnesium Trisilicate Oral Powder, BP 1988

Ingredients	Master Formula	Reduced Formula
Magnesium Trisilicate	250g	-
Chalk, in powder	250g	-
Sodium Bicarbonate	250g	-
Heavy Magnesium Carbonate	250g	
Send 20g		

Formulation: The ingredients are all ‘active’

Compounding: Mix the powders in a mortar in order of increasing bulk volume. Pass the resulting mix through a 250µm sieve lightly re-mix and pack.

Storage: store in a dry place

Container: amber glass screw-capped jar or plastic pot

Advice for patients when dispensed: The powder should be taken mixed with a little water or other fluid between meals. If counter-prescribed prolonged use should be discouraged without medical advice

Actions and Uses: Adsorbent and antacid

Dose: 1-5 grams

2. Oral Rehydration Salts, BP 1988 (O.R.S.)

Ingredients	Master Formula (Sufficient for 1 liter solution)
Sodium Chloride	1.0 g
Potassium Chloride	1.5 g
Sodium Bicarbonate	1.5 g
Anhydrous Glucose	36.4 g
Send for two separate 100ml doses of solution	

Formulation: The ingredients are all ‘active’

Compounding: Make a small excess to allow for losses in the mortar. Mix the powders in a mortar in order of increasing bulk. Pass the resulting mix through a 250µm sieve, lightly remix and pack

Storage: Store in a dry place

Container: Wrap each dose (4.04 g) in powder papers, or supply each dose in an individual amber glass screw capped jar or plastic pot

Advice for Patients when dispensed: Each dose of the powder should be dissolved in sufficient freshly boiled and cooled water to make 100ml of solution taking hygienic precautions. After reconstitution any unused solution should be discarded after 1 hour unless refrigerated when it may be kept for 12-24 hours

Actions and Uses: Rehydration and electrolyte replacement in the treatment of diarrhea

3. Zinc, Starch and Talc Dusting Powder, BPC 1973

Ingredients	Master Formula
Zinc Oxide	250 g
Starch, in Powder	250 g
Purified Talc, Sterilized	500 g
Send 15 g	

Formulation: The ingredients are all 'active'

Compounding: Mix the powders in a mortar in order of increasing bulk. Pass the resulting mix through a 180 µm sieve, lightly re-mix and pack

Storage: store in a dry place

Container: amber glass screw-capped jar or plastic pot with recloseable perforated lid

Advice for patients when dispensed: The powder should be dusted lightly on to the affected area. It should not be applied to broken skin or to raw surfaces of large area

Actions and Uses: Adsorbent dusting powder

Exercise

1. Send four powders each containing 8mg-propranolol hydrochloride for a child weighing 8kg
 - **Hint:** Use lactose as inert diluent
 -

Experiment 10

GRANULE DOSAGE FORMS

- Granules are prepared agglomerates of smaller particles. They are usually irregularly shaped and behave as single larger particles, ranging from 2 to 4 mm in diameter
- Granulation is a convenient way of dispensing solid medicaments that have very large dose and unpleasant flavor. Granulation of a powder allows addition of flavoring and coloring agents and produces an easily handled and attractive product

Compounding

- Granules are prepared on a small scale with a mortar and pestle and suitable sieves. The ingredients are first moistened with a granulating agent until they are coherent but not too damp. Suitable granulating agents include water, starch mucilage, gelatin or sucrose solutions and dilutions of alcohol. The coherent material is then pressed through a suitable sieve. The resulting granules are shaken gently on a finer sieve to remove loose powder, and then dried in a drying oven at a temperature not more than 60°C. Uniformity of size and freedom from powder ensure even dosage

Practical

1. Methylcellulose Granules, BP 1988

Ingredients	Master Formula
Methylcellulose 2500 or 4500, in powder	64 g
Amaranth, food grade of commerce	20 mg
Saccharin Sodium	100 mg
Vanillin	200 mg
Acacia, powdered	4 g
Lactose, sufficient to produce	100 g
Send 25 g	

Formulation: High viscosity grades of methylcellulose are used as intestinal bulking agents. Vanillin and saccharin are used to sweeten and flavor the granules and amaranth is a permitted color. Acacia helps to bind the granules.

Compounding: Mix the powders in a mortar in order of increasing bulk, taking care to distribute the dye evenly. Add sufficient water, in small amounts, to form a coherent mass

- **N.B.** If the mass is too wet the granules will be too large, and if too dry there will be too much un-granulated powder

Press the mass through a 2.8mm sieve with a spatula, and shake the resulting granules gently on a finer sieve (710 μ m sieve) to remove loose powder. Spread the granules thinly on a tray and dry in an oven at a temperature not exceeding 60°C

Storage: store in a cool, dry place

Container: Wide-mouth, screw-capped glass jar or plastic pot. A 5ml spoon should be supplied for measuring the dose

Advice for patients when dispensed: The granules should be placed on the back of the tongue and swallowed with water. The granules should be taken well diluted with water

Actions and uses: Intestinal-bulking agent used as a laxative

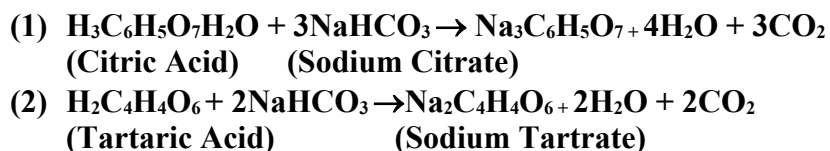
Dose: As a laxative, 1.5 to 6 grams daily in divided doses

Experiment 11

EFFERVESCENT GRANULES

- Effervescent granules are granulated mixtures of medicinal agents, often inorganic salts, with an effervescent base, which may be composed of citric acid or tartaric acid (or both) with a bicarbonate, usually sodium bicarbonate
- Effervescent salts are usually prepared from a combination of citric acid and tartaric acid rather than a single acid because the use of either acid alone presents difficulties
 - When tartaric acid is the sole acid, the resulting granules lose their firmness readily and crumble. Citric acid alone results in sticky mixtures, which are difficult to granulate
- A good working formula, for the effervescence-producing agents, has been found to be 53% sodium bicarbonate, 28% tartaric acid, and 19% citric acid
- When effervescent products are added to water, the acids and the base react to liberate carbon dioxide, resulting in effervescence and the resulting carbonated solution masks the usually saline or otherwise unpleasant taste of the medicinal agent present

- The reactions between citric acid and sodium bicarbonate (1) and tartaric acid and sodium bicarbonate (2) may be shown as follows



Methods of preparation

- Effervescent granules are prepared by two general methods: the wet method, and the dry or fusion method
- In the fusion method, the one molecule of water present in each molecule of citric acid monohydrate acts as the binding agent for the powder mixture
- The wet method differs from the fusion method in that the source of binding agent is water added to a nonsolvent moistening agent such as alcohol, which is employed to form a pliable mass for granulation
 - In the wet method, all the powders may be anhydrous so long as water is added to the moistening liquid

Practical

1. Effervescent Sodium Phosphate, USP

Ingredients	Master Formula
Sodium Phosphate, dried and powdered	200g
Sodium Bicarbonate, in dry powder	477g
Tartaric Acid, in dry powder	252g
Citric Acid, monohydrate	162g
Send 25g	

Formulation: Sodium phosphate ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$) is efflorescent in dry air. When heated to 100°C , the temperature of granulation, it loses its waters of crystallization. Failure to remove the waters of crystallization would lead to release during granulation, resulting in the acids reacting completely with the bicarbonate, and a final product that will not effervesce

Compounding: Place a large porcelain or stainless steel evaporating dish on a water bath, with as much of the dish as possible exposed to the water or steam; heat the water to boiling. The dish must be hot to ensure rapid liberation of the water of crystallization from the citric acid

- **N.B.** If heating is delayed until the powder is placed in the dish, much of the water liberated slowly as the temperature rises will be lost by evaporation

Powder the ingredients and sieve them through a $250\mu\text{m}$ sieve. Mix them in a mortar in order of increasing bulk. Place the mixed powder in the dish and press down with an acid

resistant (e.g. stainless steel) spatula until the mixture forms a loose cake or a damp coherent mass

- **NB.** The heating stage takes 1 to 5 min

Press the mass through a 2.8mm sieve with a spatula, and shake the resulting granules gently on a finer sieve (710µm sieve) to remove loose powders. Immediately spread the granules thinly on a tray and dry at not more than 54°C in a drying oven

Storage: Store in a cool, dry place

Container: An air tight, amber screw-capped jar or plastic pot with a mouth wide enough for easy introduction of 5ml spoon

Advice for patients when dispensed: About 2 teaspoonfuls of the granules are added to a glass of water, and the mixture is swallowed as the effervescence subsides

- **NB:** Each 10g of the granules contain 2g of desiccated sodium phosphate, the usual dose of this agent

Actions and uses: Cathart

Experiment 12

INCOMPATIBILITY

- Incompatibility is the result of mixing two or more antagonistic substances and it is detected by changes in the physical, chemical, or therapeutic qualities

Types of Incompatibility

- There are basically three types of incompatibility: **1.** Therapeutic incompatibility, **2.** Physical and **3.** Chemical incompatibility

(1) **Therapeutic Incompatibility:** it occurs when the response to one or more drugs in a patient is of different nature and intensity to that of the intended one

(2) **Physical Incompatibility:** it is usually due to immiscibility, insolubility and liquefaction

- **Immiscibility**

- i. Mixing oil and water in emulsions

Solution:

- **Insolubility**

- i. For solids that are sparingly soluble, in the preparation of solutions

Solution:

- ii. For solids that are poorly wettable, in the preparation of suspensions

Solution:

- iii. For indiffusible solids, in the preparation of suspensions

Solution

- **Liquefaction**

- i. Formulation of eutectic mixtures, e.g. in the preparation of powders

Solution:

- ii. Liberation of water of hydration, e.g. in the preparation of effervescent granules

Solution:

Examples

1. Concentrated peppermint Emulsion

Ingredients	Master formula
Peppermint Oil	20ml
Chloroform Water (DS)	500ml
Water to	1000ml
Polysorbate 20	1ml

Formulation: Polysorbate 20 is the emulsifying agent

Actions and uses: Flavoring agent

2. Phenobarbitone Elixir

Ingredients	Master formula
Phenobarbitone	0.4 g
Compound Orange Spirit	2.5 ml
Amaranth Solution	1 ml
Purified Water, to	100 ml
Ethanol (90%)	40 ml
Glycerol	40 ml

Formulation: Phenobarbitone is virtually insoluble in water (1 in 1000), but a clear product can be made by dissolving it in alcohol and then diluting with glycerol and water. Ethanol (90%) and glycerol are included as co-solvents for the phenobarbitone

Actions and Uses: Anti-convulsant

3. Sulfur Lotion, Compound, BPC 1973

Ingredients	Master formula
Precipitated Sulfur	40 g
Glycerol	20 ml
Alcohol (95%)	60 ml
Calcium Hydroxide Solution to	1000 ml
Quillaia Tincture	5 ml

Formulation: Quillaia tincture (0.5 per cent) is used as a wetting agent for the poorly wettable sulfur. **Note that** the saponins of quillaia in aqueous solution lower surface tension, and the extract and tincture are used as wetting agents.

Glycerol is an emollient and with the alcohol helps to wet the sulfur and promote dispersion

- **NB.** In unofficial sulfur lotion, 0.25 per cent sodium lauryl sulfate may be used instead of quillaia tincture

Action and uses: Mild antiseptic for acne and scabies

4. Phenolated Calamine Lotion, USP

Ingredients	Master formula
Calamine	80 g
Zinc Oxide	80 g
Glycerin	20 ml
Liquefied Phenol	10 ml
Calcium Hydroxide Solution to	1000 ml
Bentonite	20 g

Formulation: Calamine (basic ZnCO_3 suitably colored with Fe_2O_3) and Zinc oxide are both indiffusible solids and therefore bentonite is included as a thickening agent

Actions and uses: Cooling lotion useful for treating mild sunburn

5. Eutectic mixtures (Bulk Powder)

Ingredients	Master formula
Menthol	0.2 g
Camphor	0.4 g
Zinc oxide	6.0 g
Talc	3.4 g

Formulation: Menthol and camphor form eutectic mixtures upon intimate mixing. So the talc is used as an inert sorbent to adsorb the eutectic mixture

Actions and uses: Treatment of inflammation of sweat glands

6. Effervescent Sodium Phosphate, USP

Ingredients	Master formula
Sodium Phosphate, dried and powdered	200 g
Sodium Bicarbonate, in dry powder	477 g
Tartaric Acid, in dry powder	252 g
Citric acid, monohydrate	162 g

Formulation: Sodium phosphate ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$) is efflorescent in dry air. When heated to 100°C , the temperature of granulation, it loses its waters of crystallization. Failure to remove the waters of crystallization would lead to release during granulation, resulting in the acids reacting completely with the bicarbonate, and a final product that will not effervesce

Actions and uses: Cathartic

(3) Chemical Incompatibility

- Chemical incompatibilities are generally caused by pH changes, chemical reactions and complexations
- Chemical incompatibility may be manifested by
 - Formation of a precipitate

- Color change, or
- Gas production

Practical

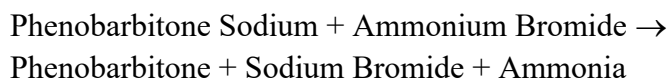
1. Private prescription

Ingredients	Master formula
Phenobarbitone Sodium	600mg
Ammonium Bromide	8g
Water to	100ml

Make a mixture, send 25 ml

Label: Two 5ml spoonfuls to be taken at night

- **Note:** Phenobarbitone sodium and ammonium bromide react to produce ammonia gas and insoluble free phenobarbitone



- However, since NH_4Br was used for the sedative action of the bromide ion, it is permissible to substitute the chemically equivalent amount of sodium or potassium bromide

$$\text{Mol. Wt. NaBr} = 102.9$$

$$\text{Mol. Wt. KBr} = 119.0$$

$$\text{Mol. Wt. NH}_4\text{Br} = 97.95$$

$$\text{Amount of KBr required} = \frac{8 \times 119}{97.95} = 9.7\text{g}$$

97.95

Similarly, the amount of NaBr required = 8.4g